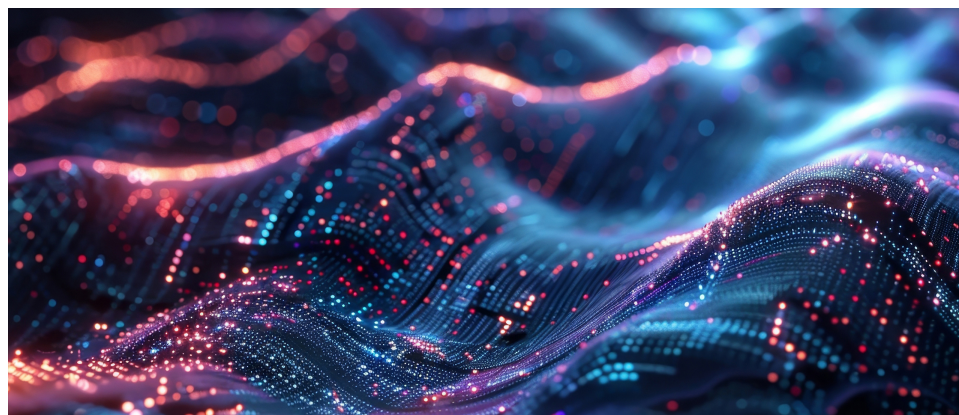


Positioning for AI Disruption

The Next Phase of Enterprise Adoption

As AI deployment scales, the industry is responding to elevated spending by controlling the intelligence stack and placing a greater focus on ROI. We assess how recent trends are influencing where value accrues, identifying software beneficiaries and 70 companies executing on AI adoption.



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- **AI economics are shaping the next phase of enterprise adoption:** As the cost of a unit of intelligence declines and AI capabilities improve, broader adoption and increasingly complex use cases are driving spending higher. In response, we see a greater emphasis on managing AI deployment, with advances in models, compute, frameworks, and routing technologies creating new opportunities to optimize cost and performance.
- **We see two key takeaways for the investment theme:** 1) software vendors will play a critical role in enterprise deployment, benefiting those who can successfully monetize AI, and 2) execution capabilities are becoming the key differentiator for companies adopting AI. We identify [companies across both buckets](#) and outline implications across the AI value chain.

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Executive Summary

As AI capabilities continue to advance rapidly, elevated investment across the AI value chain has increased scrutiny around the ROI. Since publishing **Positioning for AI Disruption (U.S. & EU) Editions** earlier this year, the **AI cost paradox** has become more apparent. The upfront cost of AI deployment remains high even as the cost per unit of intelligence declines. At the same time, broader AI adoption and the shift toward more complex and agentic workflows is driving enterprise spend higher.

In turn, **companies are responding by re-evaluating how AI is deployed from both a technology and application perspective.** The rise of highly-capable open-weight and lower-cost models paired with advancements in AI frameworks are allowing organizations to exert greater **control over their intelligence stacks**, with software vendors playing a key role in enabling this shift. **Model routing** technologies are being adopted to match workloads with the appropriate balance of performance, cost, latency, and security – creating a "best tool for the job" approach when it comes to choosing AI models.

Organizations are also moving from broad experimentation toward **more targeted deployment**, placing a greater emphasis on ROI, workflow redesign, and governance. Software vendors are responding to customer concerns by optimizing their stacks to lower AI delivery costs while also introducing more sophisticated spending controls. Further, costs are not the only consideration given rising AI safety concerns as models advance. Together, these changes influence both the timing of AI impacts and where value ultimately accrues.

Investor Guidebook: Who Captures the Value?

To help translate what this means for the AI investment theme, we map implications across the value chain and identify key companies (**Figure 1**):

1) Enterprises Executing on AI Adoption

For companies adopting AI, as access to highly-capable, lower-cost models becomes more widespread, we see model access alone becoming less of a differentiator. Instead, competitive advantage is more dependent on an organization's ability to deploy AI effectively and translate adoption into measurable business outcomes.

In our view, this **raises the importance of execution capabilities**, including proprietary data, technology readiness, and AI talent. **We identify 70 companies across sectors** that are demonstrating AI implementation progress, reporting quantifiable benefits, or possessing company-specific characteristics that may support successful execution.

2) Software Vendors Helping Bridge AI Intelligence to Enterprise Outcomes

In addition, execution will depend on the enterprise data layer, which resides across applications, systems of record, data infrastructure, and other enterprise repositories that software vendors deliver. While foundation models are becoming increasingly intelligent, they do not inherently understand a company's unique business context, workflows, and proprietary data, meaning software vendors will play a key role in enterprise AI adoption.

At a high level, we assess the impacts for software vendors across four buckets. Broadly, we see infrastructure and cybersecurity vendors as more clearly positioned to benefit from accelerating AI adoption, with application software and vertical software to benefit in aggregate but with greater dispersion amongst winners and losers. Infrastructure software vendors provide the rails, integration, governance, and orchestration needed to connect models with enterprise context, while cybersecurity vendors are seeing growing demand to secure AI applications, data, identities, and workflows. Application vendors are embedding AI directly into their offerings, creating incremental value by allowing customers to generate insights and automate work within existing workflows, while large systems of record benefit from powerful data gravity that makes them natural platforms for enterprise AI. Taken together, **software vendors sit at the center of how enterprises operationalize AI**, which we believe positions the sector as a key beneficiary of long-term AI investment, although economics will not be shared equally and there is risk of disruption for vendors that are unable to pivot.

3) Implications Across the Intelligence and Infrastructure Layers

We also see trends in enterprise adoption and AI economics driving broader impacts across the intelligence and infrastructure layers of the AI ecosystem. The **major AI labs continue to push the frontier with more capable and efficient models**, supporting new use cases and continued growth despite rising competition from open-weight alternatives. At the same time, ongoing reductions in the cost per unit of intelligence are increasing the number of AI deployments capable of generating attractive returns. In our view, these dynamics should support **AI cloud infrastructure providers**, including **Amazon, Microsoft, Google, Oracle, CoreWeave** and **DigitalOcean**, as demand for AI compute continues to outstrip supply.

Importantly, **AI infrastructure providers are largely model agnostic**, benefiting whether workloads run on proprietary models or open-source alternatives. As a result, even if the market adopts a "barbell" approach around AI model adoption, with simpler tasks going to smaller, cheaper models while more advanced reasoning and use cases stay with the frontier models, we expect demand for the underlying compute, power, and infrastructure layers to remain strong. However, over the long term, we continue to monitor the pace of efficiency gains and company-level dynamics for the cloud service providers, such as capacity expansion, customer diversification, and balance sheet management.

FIGURE 1. Key Company Mentions

Software	
Software Vendors	
Horizontal Application Software	Atlassian, Braze, Five9, Gitlab, Intuit, Microsoft, monday.com, Oracle, Salesforce, SAP, ServiceNow, Workday
Infrastructure Software	CoreWeave, Datadog, DigitalOcean, Dynatrace, Elastic, IBM, Microsoft, MongoDB, Oracle, Snowflake, UiPath, WhiteFiber
Cybersecurity	Palo Alto Networks, CrowdStrike, Cloudflare, Varonis, Okta
Vertical SaaS	Autodesk, Procore, Veeva
AI Cloud Service Providers	Alibaba, Amazon, CoreWeave, DigitalOcean, Google, Microsoft, Oracle, WhiteFiber
AI Adopters	
Industrials & Materials	Thomson Reuters, C.H. Robinson, GXO Logistics, Carvana, OPENLANE, ACV Auctions, Republic Services, ACS, Vinci, Verisure, Rentokil, Experian, DSV, Airbus, Safran
Energy & Utilities	SLB, bp, TotalEnergies, Técnicas Reunidas, Viridien, Maire
Financials	Bank of America, JPMorgan Chase, First Horizon, Associated Banc-Corp, Royal Bank of Canada, TD Bank, Banorte, Arthur J Gallagher, The Hartford Group, Rocket Companies, Robinhood, Santander, ING, LSEG, Man Group, AXA, Allianz, Unibail-Rodamco-Westfield
Consumer	The Coca-Cola Company, Home Depot, Walmart Inc., Wayfair Inc., Ralph Lauren, Tapestry, Ulta Beauty, Royal Caribbean Group, Arca Continental, Ingredion, L'Oréal, Reckitt, Unilever, Heineken, IHG, Accor, Compass, Inditex, Next, Dunelm, Victorian Plumbing, JD Sports, Frasers Group, Tesco, Ahold Delhaize
Healthcare	UnitedHealth Group Inc, HCA Healthcare Inc, Eli Lilly, Merck, Moderna, Roche

AI Adopters reflects a screen building on our Positioning for AI Disruption deep dive - see our [U.S.](#) & [EU](#) Editions for additional detail. We highlight companies demonstrating AI implementation progress, reporting quantifiable benefits, or possessing company-specific characteristics that may support successful execution. For additional detail on the companies in Figure 1, see our Investor Guidebook.

Source: Barclays Research

UNDERSTANDING THE AI COST PARADOX

In the initial stages of AI deployment, the focus was primarily on the capabilities of the technology, with AI labs competing to build the highest-performing models and seeking to maximize the distribution of AI into the market. Excited by the potential for AI to fundamentally change business models and improve efficiency, the initial strategy for many of the consumers of AI was aggressive experimentation and driving broader adoption. While both of these dynamics remain relevant today, the rapid acceleration of enterprise spend has drawn more attention to the cost of delivering those capabilities, encouraging advances that maximize intelligence per watt or dollar spent. Importantly, pricing AI based on token consumption means there is more of an explicit marginal cost compared to other technologies that have more of a fixed cost (e.g., seat-based revenue models), elevating the importance of ROI.

Factors Driving the Cost of Intelligence Lower

It has long been understood by economists and investors that new technologies tend to see costs fall over time, and AI intelligence is no different. However, we see several important nuances here. 1) Although token prices have fallen significantly since the launch of GPT-4, pricing continues to vary widely across providers. This has resulted in a broad spectrum of price-performance options, with increasingly capable low-cost models at one end and premium reasoning models at the other (Figure 2). 2) Usage has also shifted toward lower-cost models, with recent declines in the Silicon Data LLM Token Expenditure Index reflecting changes in both pricing and model mix (Figure 3). Here, we note usage and spend/monetization are two related, but separate dynamics. 3) Interestingly, GPU rental costs (measured by GPU/hour rate) are holding firm and in many cases have increased, even as the cost per unit of intelligence declines and new generations have emerged, highlighting the continued supply/demand imbalance (Figure 4).

FIGURE 2. Token Pricing Varies Widely Across AI Models

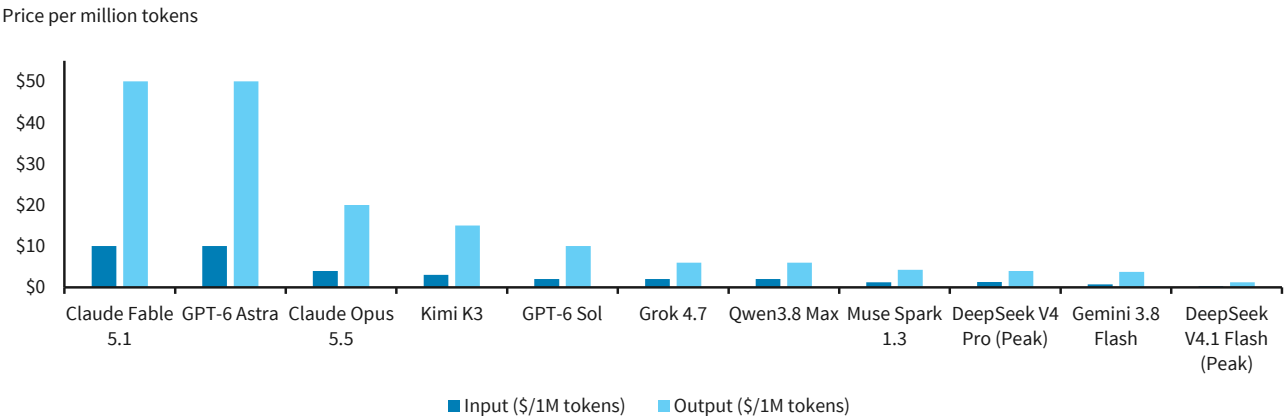
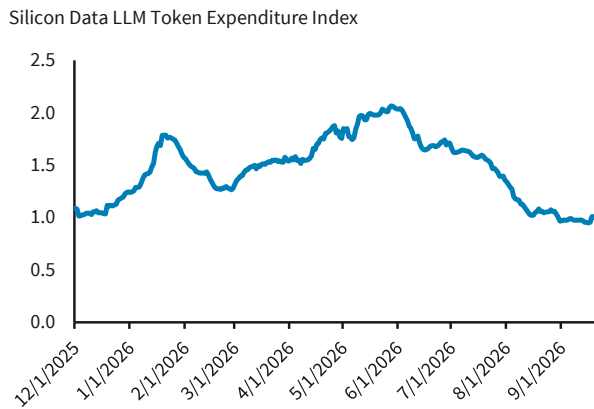
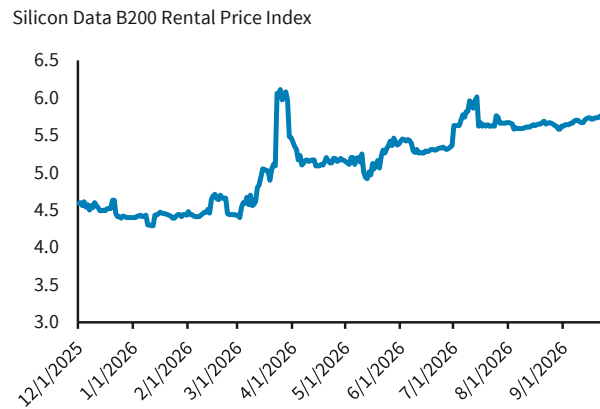


Chart represents a selection of AI models with pricing as of September 23, 2026. Gemini 3.8 Flash pricing refers to introductory pricing (through December 31, 2026). Source: Company websites, Barclays Research

FIGURE 3. Silicon Data LLM Token Expenditure Index

The LLM Token Expenditure Index is a daily statistical benchmark designed to measure the effective expenditure level of the actively traded broad LLM market, measuring price per million tokens. The index blends active provider pricing with real consumption volume observed across multi-provider routing gateways and inference networks. Pricing pulled on September 23, 2026.

Source: Bloomberg

FIGURE 4. Silicon Data B200 Rental Price Index

The B200 Rental Price Index (SDB200RT) is the Silicon Data benchmark for NVIDIA Blackwell B200 GPU rental pricing. It is published daily as a standardized, like-for-like hourly rate expressed in USD per GPU-hour. Pricing pulled on September 23, 2026.

Source: Bloomberg

Hardware, Compute, and Model Advances: Major Factors Lowering the Cost of Intelligence

While more powerful AI accelerators and supply chain constraints can increase the upfront costs of AI deployment, advances in hardware, compute infrastructure, and model efficiency are driving greater intelligence per watt.

Hardware and Compute Advancements

The semiconductor layer remains one of the primary drivers of AI infrastructure efficiency, with each new hardware generation delivering materially better performance per watt through advances in process technology, memory architectures, interconnects, and system-level design. At the same time, power densities continue to rise as the industry shifts from individual accelerators to rack-scale systems, requiring increasingly sophisticated liquid cooling, power delivery, and thermal management. NVIDIA's NVIDIA Vera Rubin NVL72 platform exemplifies this trend, with the company claiming up to 10x higher inference performance per watt than its Blackwell predecessor and up to 4x higher training performance per watt through tighter CPU-GPU integration, HBM4 memory, next-generation NVLink fabrics, and rack-scale optimization. While rack-level power consumption is increasing substantially, the corresponding gains in throughput are rising even faster, improving the amount of AI work delivered per unit of power and enabling operators to generate significantly more tokens, training output, and inference capacity within constrained data center power budgets.

Leading silicon providers improve the performance of their systems each generation on an approximately one-year cadence that drives better TCO and tokens/watt. With the growing diversification of models and AI workload types, we are starting to see more specific silicon designed for specific tasks. The first iteration came in custom ASICs with some designed specifically for training and some for inference with certain requirements. We think we will see more disaggregated inference, where different hardware is used to run compute intensive prefill vs. memory bandwidth intensive decode within inference. This can be taken a step further where different types of accelerators are used to run different steps within decode such as Attention and the Feed Forward Network (for more thoughts see our Hot Chips takeaways [here](#)). There are many benefits to disaggregated inference with AMD and CBRs pointing to 5x TPS/kW on a Helios + Cerebras infrastructure vs. standalone Cerebras. The importance of silicon is

expanding beyond the accelerator providers where total system performance is now the new metric with interconnect and memory architectures influencing designs for different types of workloads.

Custom silicon has always been an avenue for architecting hardware to match specific workload types and now even model types. GOOGLE is [reportedly](#) working on a project called "Frozen", where they design parts of their Gemini model into the silicon which could improve efficiency 6-10x. AMD also recently acquired Taalas, which uses the same principal of encoding a model into the chip. We view this as one of the more extreme examples of codesign where you are locked into a single model or model architecture. While these chip designs can provide performance improvements, they can be limited in scope, take a while to come to market, and are at risk of obsolescence if model architectures or requirements change. On the less extreme side, chips codesigned for models (OpenAI's Jalapeño) will bring more efficiencies to the market when they are ramped at scale.

Additionally, the compute layer, broadly characterized by the hyperscalers and neoclouds (discussed more within), is also optimizing to deliver better performance per watt primarily through software efficiencies and network design. The mechanisms here are rather technical, but can be more easily understood by working backwards from the sought after end result, which is essentially to keep the GPUs/AI accelerators as "busy" as possible and reduce bottlenecks between the accelerators and other hardware/networking components. Given compute performance for real life workloads virtually never sustain the theoretical maximum compute throughput, driving better utilization yields better overall performance without changes in the underlying accelerator.

Players in the compute layer accomplish this through improved techniques such as workload scheduling, batching requests, autoscaling, smarter request routing, and moving models dynamically between machines:

- **Workload scheduling:** Workload scheduling is becoming an important lever for driving better performance as AI cloud service providers continuously handle jobs with different requirements, such as training jobs, real-time or batch inference, model embeddings, and model fine-tuning. Workload scheduling helps choose where and when the workload runs, providing opportunities to more efficiently match workloads with available resources.
- **Batching:** AI cloud service providers receive billions of queries each day, which given the parallel processing capabilities of modern accelerators, gives an opportunity to group or "batch" requests together. Microsoft [noted](#) that hyperscalers such as Microsoft Azure can process thousands of requests at the same time and that usually, the bigger the system, the more efficiency improvement compounds.
- **Autoscaling:** A related but distinct function to workload scheduling that "scales up" or "scales down" resources assigned to a workload as demand changes. Scaling can be done vertically (more/less CPU or GPU resources) or horizontally (adding/removing servers).
- **Smarter request routing and moving models dynamically between machines:** Innovations have allowed cloud service providers to route requests to different hardware and models, which can help match workloads to the most efficient resources. Additionally, models can be moved based on demand, which Google [noted](#) is a lever it uses to drive better efficiency.

Some other notable techniques include optimizing the data center itself (e.g., cooling efficiencies), disaggregating prefill (read/processing prompt) from decoding (generating tokens), which then can be optimized independently; quantization (in layman's terms,

representing numbers with fewer bits. Note, quantization also involves the hardware/model layer), which lowers the memory requirement but can come with performance trade-offs; and speculative decoding (generating multiple tokens at once using smaller models that "guess" a sequence of tokens that gets verified by a larger model).

Model Advances

We note that hardware and compute advancements are not the only levers driving better AI cost economics, with the large AI model labs also continuing to drive better performance through engineering optimizations and efficiencies. This includes architectural innovations such as Mixture of Experts (MoE), which activate only a subset of parameters during inference, as well as training techniques such as distillation, where a smaller model is trained to replicate much of the capability of a larger one, and inference optimization. Together, these advances can compound, driving significant gains in model efficiency. For example, according to OpenAI (private, not covered), GPT-5.6's "efficiency edge comes from improving the models, the inference systems that run them, and the agentic harness that connects them to tools and context". We expect continued progress in this area with future model generations.

We also note new AI model approaches are emerging, which could impact token economics for certain use cases. Below, we discuss Jev, a new model from TypeSafe AI that takes a different approach from traditional LLMs, as a case study to highlight continued innovations in the model layer.

Case Study: Jev

We also acknowledge that the AI model ecosystem is evolving rapidly, with new types of models being developed that are optimized for different use cases. For example, TypeSafe AI (private, not covered - start-up founded by former OpenAI researcher Diogo Almeida) released a new model last week called Jev, which takes a different approach from traditional LLMs. Rather than generating text for human consumption, Jev is designed to make fast and structured decisions useful for software and AI agents, improving AI unit economics through lower latency and token costs. Within, we provide background on the model's "System One" architecture, explain how Jev differs from traditional LLMs, and discuss the potential broader implications for agent economics, enterprise adoption, and the AI ecosystem.

What is Jev? Jev is TypeSafe AI's first "System One" model, aimed at making fast, structured decisions to give software/agents optimized outputs while removing text generation meant for humans as the end-user (System Two models). TypeSafe AI argues that many agentic workflows involve repetitive decision-making tasks such as ticket routing, tool selection, and urgency assessment, which can make text generation unnecessarily expensive and slow. When reasoning, the model uses parallel sampling (as opposed to sequential) to generate all outputs simultaneously, instead of one at a time. This approach allows for lower latency and token costs compared to System Two models. The company claims Jev can operate with an end-to-end response time of 70-500ms and can be 40-200x faster than frontier models. Jev also uses a training method called Reinforcement Learning for Calibrated Decisions (RLCD), lowering risk of model hallucinations and type errors. After its release last week, Jev has already been integrated as an API into Vercel, Cloudflare, LangChain, and Langfuse.

- **Improved agent economics allow for cheaper pricing in comparison to frontier models.** Jev is currently priced at \$0.042 per million input tokens, as opposed to \$0.20 to \$10 for existing LLMs, as quoted by TypeSafe AI. Given Jev avoids text generation, it does not charge for output tokens and cites that existing LLMs charge a ~5x higher rate for their output tokens compared to their input token costs. An independent test from Every ([see here](#)) resulted in ~580x cheaper performance on extraction tasks, but we note this can vary heavily by task.
- **Accuracy remains an area to be monitored.** While the stripping away of text generation can lead to fewer hallucinations and type errors, Jev is not necessarily more intelligent than frontier models. TypeSafe AI's internal evaluation showed Jev achieving ~67.8% accuracy in comparison to ~74.1% accuracy for the best competitor model. This is a meaningful 6 percentage point gap, showing the trade-off between price and quality, but we note top-model intelligence is not the value proposition of Jev.
- **Jev is not a replacement to frontier models, but a complement.** Jev is meant for software-to-software communication, meaning it should be used for structured decision-making tasks such as routing, classification, tool selection, and workflows. Frontier LLMs remain better suited for reasoning, content generation, coding, and human-facing interactions.

Intensifying Competition: The Rise of Lower-Cost Alternatives

Another important dynamic influencing the cost of deploying AI is the rise of open-weight models, which are often free to download or can be often accessed at lower per-token rates. The performance gap between open and closed models has narrowed, intensifying competition as a broader range of models become capable of meeting the performance requirements of many tasks.

In particular, the launch of highly-capable Chinese models, such as DeepSeek, Qwen, and Kimi, have expanded the landscape of alternatives to U.S. frontier models. An expert hosted by Barclays in July 2026 [shared his view](#) that the performance gap between the U.S. and China has compressed from roughly two years initially, to six months, to potentially six weeks following the release of Kimi K3. However, we note this performance gap is constantly changing as new models are released from the frontier labs.

This has raised discussion around the strategic importance of open models. In China, support for open models has been an important component of the country's AI strategy (see our [case study](#)), while in the U.S., interest in open models has become more visible in recent months. For example, a [letter](#) signed by a group on tech companies, including NVIDIA, Microsoft, and Meta, expressed the importance of open-weight models in making "advanced AI more accessible, adaptable, and widely available".

Model Mix is Shifting, Putting Pressure on Pricing

As open-weight models have become more capable, they have started capturing a growing share of inference demand. Data from both OpenRouter and Vercel indicates that a rising share of tokens is being routed to open-weight models. An important nuance, however, is that share of tokens and share of spend can differ significantly given the difference in pricing. For example, according to Vercel, open-weight models ran 56% of gateway token volume in August 2026, but accounted for only 14% of spending.

Increasing competition has also been reflected in recent pricing updates and model releases. For example, OpenAI launched GPT-6 with 50% lower pricing than GPT-5.6 Sol and Luna. Similar trends have been seen across the industry. Google introduced Gemini 3.7 Flash at half the original 3.6 Flash cost per million tokens just three weeks after Gemini 3.6 Flash launched. Anthropic (private, not covered) recently launched Claude Opus 5.5, which costs 40% less to run than Opus 5. Hence, we expect key players to continue to come out with different models catered to varying performance/cost requirements to maximize the workloads they are able to economically address.

As we detail in our section on [Controlling the Intelligence Stack](#), the growing competitiveness of open-weight models is creating opportunities for intelligence providers and enterprises to reduce AI costs through model selection. However, as Chinese models gain market share, this has also called attention to the implications for U.S. competitiveness and raised discussion on potential policy actions.

Balancing Two Key Considerations: Sovereign AI and Safety

Policymakers are increasingly balancing concerns around safety with efforts to strengthen domestic AI. Debate on appropriate safeguards has escalated amid rapid advances in model capabilities and growing cybersecurity concerns. For example, Anthropic CEO Dario Amodei proposed to slow the pace of frontier AI development in a September 12 letter, a position supported by OpenAI's Sam Altman, Google DeepMind's Demis Hassabis, and xAI's Elon Musk, but dismissed by President Trump. At the same time, Amodei called for policies to defend the U.S.'s AI advantage, including strengthening chip bans and cracking down on unauthorized distillation. In our view, at a high level, it will be relevant to monitor AI policy on two related fronts: 1) efforts to support the safe deployment of increasingly advanced AI models, and 2) measures targeting domestic capabilities and competitiveness.

Although much of the attention has been on the U.S. and China, [strategic autonomy](#) has also emerged as a key theme in Europe, including an increased focus on AI. The recent State of the Union (SOTEU) address by the President of the European Commission highlighted the importance of Europe having its own capabilities. This has also been reflected in private markets. Earlier this month, Mistral raised €3bn in a Series D funding round, valuing the company at more than €21bn. Mistral is aiming to advance AI sovereignty with its full-stack AI solutions, including open-weight models, and plans to build up to 1 GW of capacity across the EU by 2030.

Special Report: China Case Study

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This section is a Special Report that is not an equity or a debt research report under U.S. FINRA Rules 2241-2242.

Case Study: China AI Adoption Roadmap: Open Models + Wide Industry Adoption

China's AI push dates back to 2016, when its first AI policy was launched under the Internet+ initiative¹, coinciding with AlphaGo's landmark 4-1 victory over Go champion Lee Sedol. In August 2025, Beijing unveiled the nationwide AI+ Action Plan, setting AI adoption targets for intelligent terminals and AI agents of 70% by 2027 and 90% by 2030. The plan prioritizes six areas: science and technology, industry, consumption, public services, governance, and international cooperation. In March 2026, the initiative was formally embedded in the 15th Five-Year Plan (2026-2030).

Following the national blueprint, government agencies including the MIIT, MOFCOM and NEA have rolled out sector-specific policies across the six priority areas. Within industry and consumption alone, we identify 9 industry groups and more than 100 targeted sectors and business activities². Notably, only a subset of these sectors is tied to quantitative targets, while many are referenced as illustrative use cases or deployment scenarios.

Model developers migrating to more open weights

While China does not explicitly discourage closed-source models, the term "open model" appears nine times in the AI+ Action Plan, particularly under the AI + Global Cooperation pillar. By contrast, references to closed-source models are largely absent from both national and industry-level policy documents. At the local level, Zhejiang has set a target of fostering more than 1,000 open-model ecosystem companies by 2026.

Coupled with the government's promotion on open models, we also observe growing "openness" in the latest model launches across a number of developers. Since 2024, many developed have shifted towards Apache 2.0 or MIT licenses, which allow weight download, fine-tuning, commercial usage, modification and self-services.

A clear inflection point came with DeepSeek R1, which combined frontier-level performance with fully open weights under an MIT license. Shortly thereafter, Moonshot AI made a dramatic shift in strategy, releasing Kimi K2 under a modified MIT license³ in July 2025 and moving away from its previous API-centric approach. Baidu, after years of advocating closed-source development, also pivoted by open-sourcing the entire ERNIE 4.5 model family under Apache 2.0 in early 2025.

¹ "Internet+ and Artificial Intelligence Three-Year Action and Implementation Plan", National Development and Reform Commission, Ministry of Science and Technology, Ministry of Industry and Information Technology, Cyberspace Administration of China, May 2016

² Manufacturing, Energy, ICT, Transportation, Agriculture, Consumption Goods, Robots, Services and Commercial

³ Kimi's modified MIT: If a company or product exceeds certain scale thresholds (reported as >100 million monthly active users or >US\$20 million in monthly revenue), it must prominently display attribution to Kimi in its product. In simple terms, large companies cannot use the model invisibly; they must acknowledge that Kimi technology is being used.

FIGURE 5. Licensing Changes for Chinese AI Models

Developer	2023	2024	2025	2026
Deepseek	Open weights	Open weights	MIT	MIT
Kimi	Closed	Closed	Modified MIT	Modified MIT
Alibaba	Custom	Mixed	Apache	Apache
Zhipu AI	Open	Open	MIT	MIT
Baidu	Closed	Closed	Apache	Apache
Tencent	Closed	Community	Community	Apache

Source: Company data, Barclays Research

(i) This author is a debt research analyst in the Fixed Income, Currencies and Commodities Research department and is neither an equity research analyst nor subject to all of the independence and disclosure standards applicable to analysts who produce debt research reports under U.S. FINRA Rule 2242.

Enterprise Spend Is Rising Despite Cost Declines

For end users, total AI spend is driven by the cost of intelligence, how many tokens each task consumes, and how many tasks are run. This means token pricing tells just one part of the story. We see cost per task becoming a more relevant metric than cost per token for many enterprises.

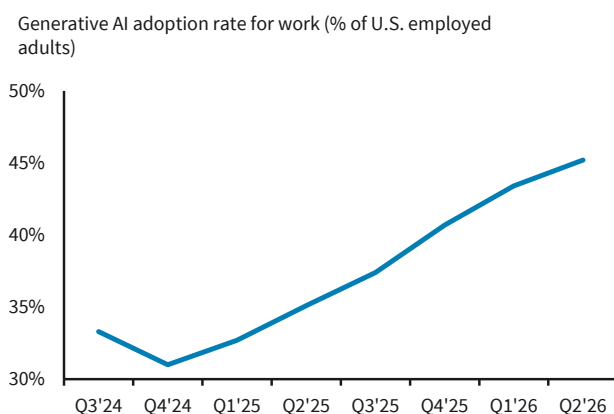
In some cases, a model with a higher price per token may complete a task more accurately or with fewer tokens, resulting in a lower overall cost. Cost per task also captures ongoing changes in AI workloads. As enterprises move toward more complex and agentic applications, the amount of intelligence required to complete a workflow is evolving. In turn, enterprise AI spend is shaped not only by the expansion of AI adoption across users and use cases, but also by changes in how AI is being used.

AI Adoption Is Maturing

Enterprise AI adoption is broadening across both users ([Figure 6](#)) and use cases ([Figure 7](#)), signaling deployment is maturing. Insights from our sector analysts suggest AI adoption is starting to progress beyond the pilot phase, as companies begin to quantify the benefits and set strategic targets. Additionally, advances in frameworks, harnesses, governance and security play an important role in this movement beyond the pilot phase. More than 50% of our sector analysts expect AI will drive moderate to significant benefits for their sector in the next 1-2 years. Importantly, this extends beyond companies exposed to the AI infrastructure build-out to those integrating AI into their own operations across industries such as healthcare and financials, as detailed in our [Barclays AI Analyst Sentiment Heatmap](#).

Expectations for AI to drive incremental revenue, cost savings, and productivity gains has translated to higher spending. According to [Ramp](#), an enterprise spend-management platform, AI token spend increased 572% from June 2025 to June 2026. While broader AI adoption remains a key driver of this dynamic, it is not the only factor. Users are moving toward more sophisticated AI applications, increasing token consumption per task. Agentic AI is also expanding the types of tasks that can be performed, enabling more complex and higher-value workflows.

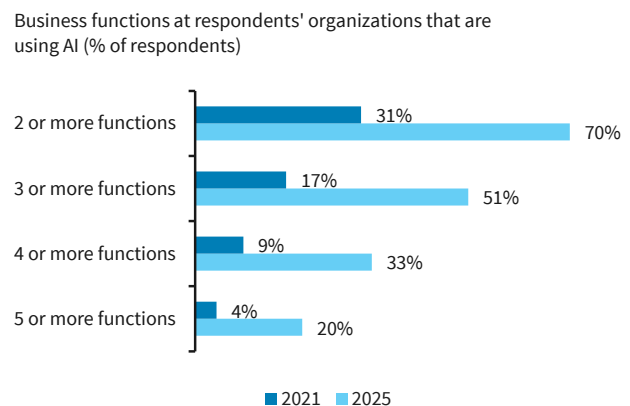
FIGURE 6. Generative AI Adoption for Work Is Rising



Share of employed adults aged 18-64 who use generative artificial intelligence for work

Source: The Rapid Adoption of Generative AI, A. Bick, A. Blandin, and D. Deming, Management Science (2026) (<https://pubsonline.informs.org/doi/10.1287/mnsc.2025.02523>).

FIGURE 7. Organizations Are Increasingly Using AI Across Multiple Functions



2025 data as of July 2025

Source: McKinsey, Barclays Research

More Complex Workloads Often Require More Tokens

As AI adoption matures, usage is evolving beyond simple prompts. More complex tasks can require more tokens, which can translate into higher AI costs but also greater enterprise value. Longer prompts, larger context windows, and increased use of reasoning models have contributed to higher token consumption ([Figure 8](#)). Average prompt token length has grown nearly 4x since early 2024 and reasoning models now represent half of all usage, according to OpenRouter's [State of AI](#). The shift toward agentic AI has also emerged as a key driver of token consumption, as multi-step workflows can require substantially more tokens per task.

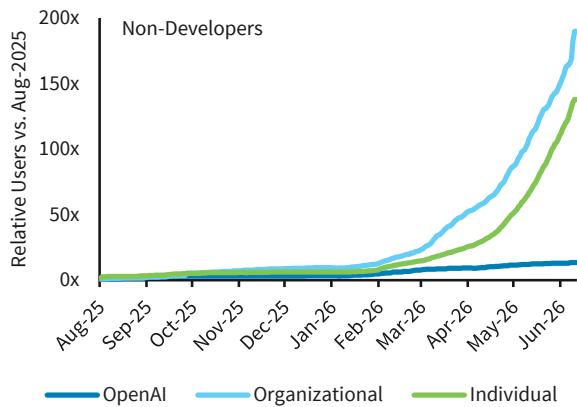
Agentic AI 101: Why are token counts higher? Agentic systems use multi-step reasoning to break tasks into a series of actions. As agents plan, retrieve relevant context, call tools, analyze results, and retry when needed, each additional model call consumes tokens, driving higher token usage per task.

Shift to Agentic AI Is Underway

Data from [OpenAI](#) and [Anthropic](#) suggests that as agentic AI capabilities improve, the work that is being done with AI is changing. OpenAI's Codex data reflects rapid acceleration in non-developer adoption, supported by the release of more user-friendly desktop and mobile applications ([Figure 8](#)). Anthropic's Claude Code data similarly points to an increase in non-coding work, with a growing share of sessions spent operating software, analyzing data, and writing documents ([Figure 9](#)). This aligns with analysis from our Thematic FICC Research team, which suggests demand for AI skills is becoming less concentrated in technical roles and increasingly embedded across the broader workforce (see [AI in action: The market for skills](#)).

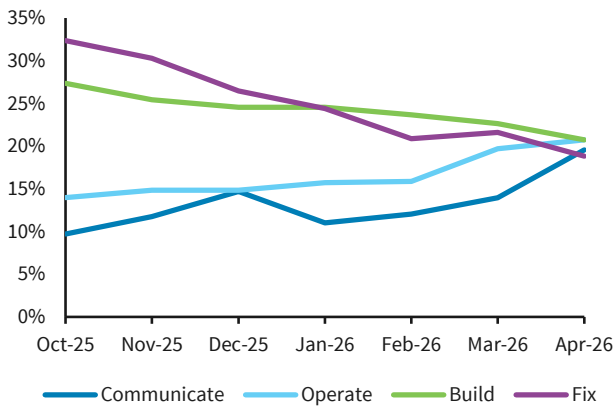
At the same time, users are trusting agents to execute more complex tasks. Codex data through May 2026 shows around 70% of sampled users made at least one request estimated to exceed 1 hour of human work and around 25% had made a request estimated above 8 hours. As AI applications move beyond simple tasks, we are seeing a shift toward higher-value workflows. Anthropic estimates the typical Claude Code task has become ~25% more valuable on average between October 2025 and April 2026, based on benchmarking to the freelance marketplace. From a cost perspective, however, as agents autonomously execute multiple model calls to complete increasingly sophisticated tasks, AI spending has grown faster than some organizations initially anticipated.

FIGURE 8. Non-Developer Adoption of Codex Has Increased Significantly in Recent Months (Aug 2025-June 2026)



Source: OpenAI, Barclays Research. See [U.S. Internet: Charts Worth a Look: Codex Quickly Becoming The Next Growth Leg At OpenAI](#)

FIGURE 9. Non-Code Work Is Making up a Greater Share of Claude Code Sessions (Oct 2025-June 2026)



Source: Anthropic, Barclays Research. See [U.S. Internet: Charts Worth a Look: Claude Code Data Shines Light on How We Use AI](#)

INDUSTRY RESPONSE: MANAGING AI SPEND

Even as the cost of intelligence falls, enterprise AI spending is outpacing those declines. Much of this spending is drawn against relatively fixed budget pools, leading to changes in how companies deploy AI. The broader software developer ecosystem has emerged as a key enabler of this shift, as organizations respond on two fronts: technical levers, such as application design, model selection and routing, and organizational ones, with a growing focus on ROI, workflow redesign, and governance.

Controlling the Intelligence Stack: AI Frameworks, Model Selection & Routing

The rise of low-cost, highly-capable models has unlocked new avenues for enterprises to reduce AI spend. However, models themselves do not translate to production AI applications and use cases. Advancing AI frameworks and harnesses help address this gap and make it easier to build AI applications and agents. In turn, companies and vendors providing software solutions now have more optionality to control their intelligence stack by leveraging emerging frameworks and harnesses, open-weight models, and routing systems. Instead of defaulting to the most capable (and in turn, often more expensive) model, enterprises can choose the model that is best suited to their use case more easily and importantly, without disrupting the underlying application.

As enterprises scale from individual AI applications to portfolios of agents operating across business functions, the strategic control point increasingly shifts from the model itself to an enterprise control layer. This layer governs identity, permissions, data access, policy enforcement, approvals, monitoring, evaluation and auditability across multiple models and harnesses. In this architecture, models become increasingly interchangeable sources of intelligence, while the enterprise control layer determines how that intelligence can be accessed and applied within enterprise workflows (see [AI Frontiers: The AI Compute Portfolio](#), 9/15/26).

Advancing AI Frameworks and Harnesses Are Making AI Development Easier

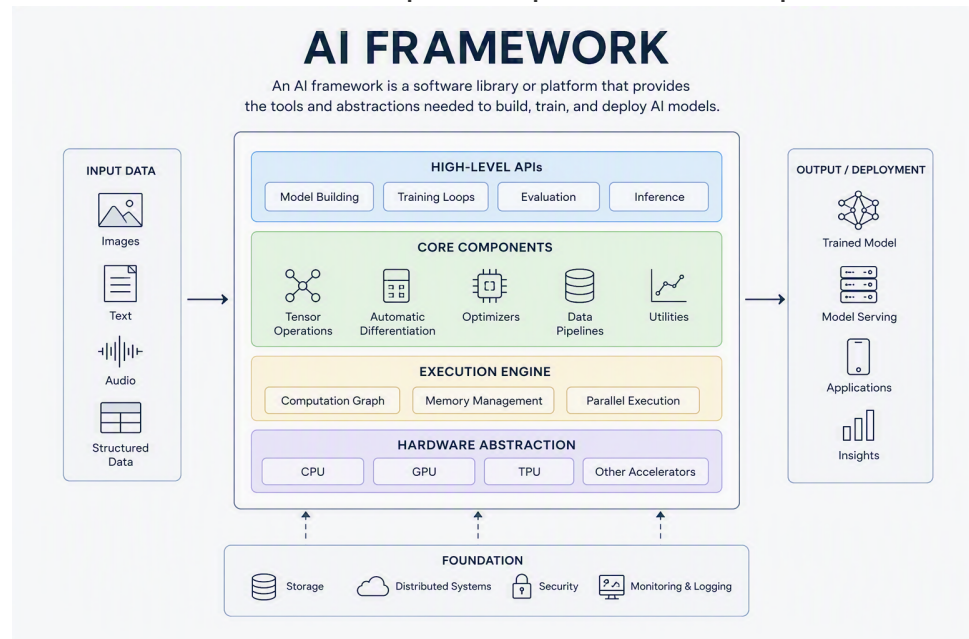
A fundamental concept for investors to appreciate is that despite LLMs continuing to improve in capabilities, these models do not inherently know a company's business, data and workflows that form the essential "context" to accurately and effectively solve many enterprise use cases. Further, these LLMs do not have native access to the software applications, tools and data repositories that employees regularly leverage to complete business workflows. Another factor is these powerful LLMs do not know the nuanced "rules" that need to be followed for a specific business task. Hence, organizations and vendors delivering AI solutions must bridge this gap between model intelligence, business context and tools of action - creating challenges for developers. However, the developer community has come together to create frameworks that are designed to make working with these AI models and building AI applications easier. In our view, advancing AI frameworks and harnesses paired with improving open-source model capabilities are accelerating AI adoption and will have implications on how AI economics are distributed amongst the ecosystem.

What Is an AI Framework?

Artificial Intelligence frameworks is a general term but in the context of AI development and this report refers to a collection of software that provide pre-built building blocks that make it easier to build, train and deploy AI and ML models. Specifically, IBM notes AI frameworks consist of datasets, libraries, packages and tools for building AI systems and early AI frameworks emerged

within academic and research communities as a way to support the development of neural networks. New frameworks were created to more easily enable specific AI use cases (e.g., natural language processing, model training, agentic AI) as developers sought to simplify the increasingly complex process of building applications on top of ML and, more recently, LLMs.

FIGURE 10. AI Frameworks Provide Developer with Components to Make AI Development Easier



Source: Barclays Research

Types of AI Frameworks

In our view, AI frameworks are best understood from the perspective of the use cases that these frameworks seek to help address. We discuss the major categories of AI frameworks below.

- 1. Assisting Model Training:** Early frameworks such as TensorFlow (Google) and PyTorch (Google) primarily abstracted the mechanics of training and executing neural networks, aiding model training. These frameworks provide pre-built components and libraries and support distributed training and hardware optimization capabilities to accelerate model development and experimentation. Other examples include NVIDIA NeMo, JAX (Google), Keras, MXNet and Hugging Face Transformers.
- 2. Streamline AI Deployment:** A second class of AI frameworks provide APIs, infrastructure components (e.g., containers) and tooling to help simplify model packaging, serving and scaling. An example of this would be tooling that helps enable multi-cloud AI deployments. Examples include vLLM, Ray Server, KServe, TensorFlow Serving, Ollama and NVIDIA Triton Inference Server.
- 3. AI Workflow and Data Management:** From a higher level, these frameworks can serve as the backbone of an AI application and help orchestrate the data pipelines, prompts and memory chains that help elevate powerful AI models to support AI agents and complex workflows. Examples include LangChain, LlamaIndex, Semantic Kernel (Microsoft), CrewAI, DSPy, and later agent-oriented systems such as LangGraph and Microsoft AutoGen.
- 4. AI Governance and Safety:** These frameworks come into play towards the end of the AI application lifecycle but serve as an increasingly important feedback loop for ensuring performant and secure AI applications. Functions span monitoring, evaluating and

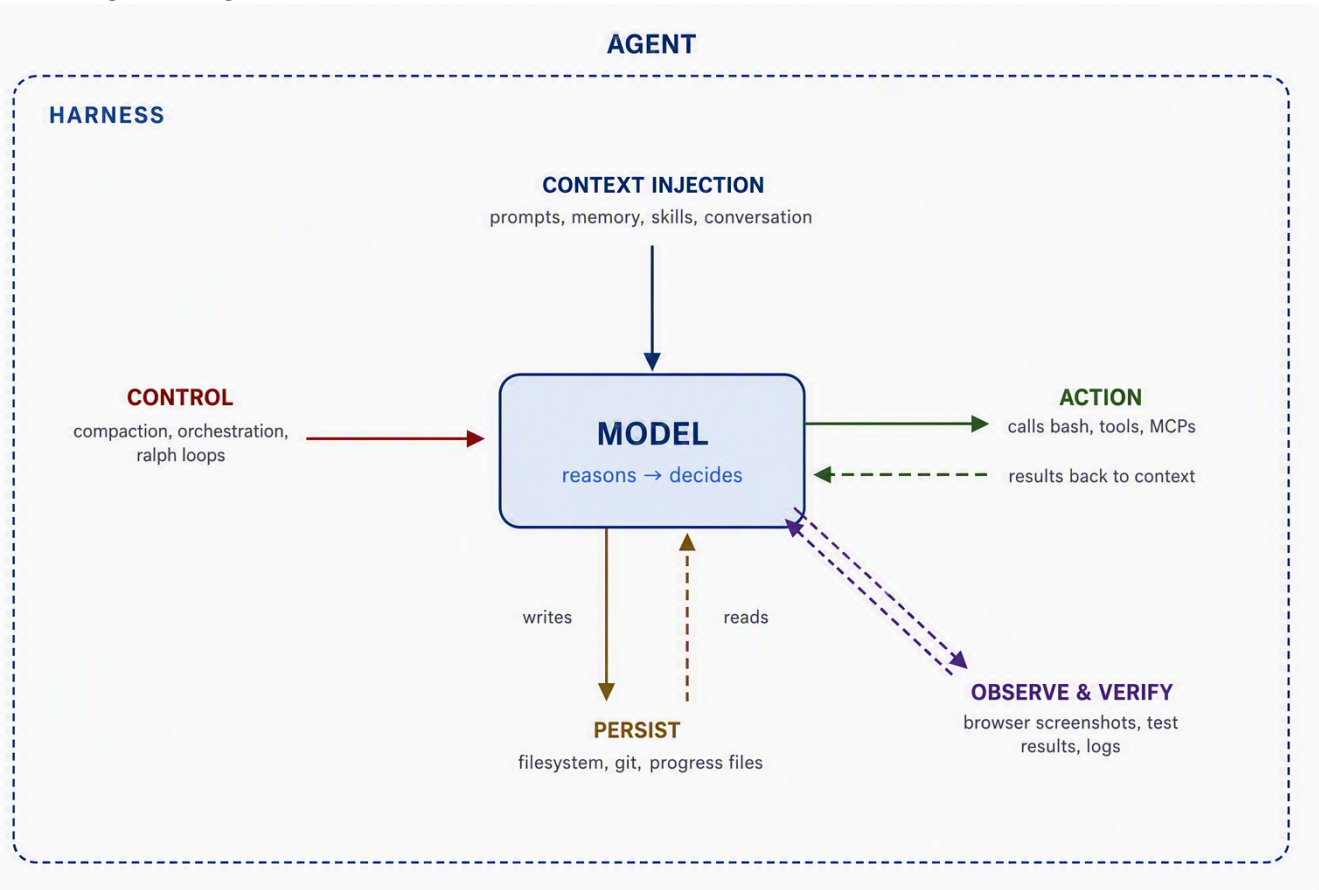
observability, enforcing guardrails and identity access. Additionally, these frameworks provide hallucination detection, bias testing and auditability.

What Is an AI Harness?

AI harnesses, often referred to as agentic harnesses, draws many comparisons to AI frameworks and are often built leveraging AI frameworks, but typically have an important nuance. We note this area is seeing rapid innovation and definitions are evolving, but Databricks eloquently defines an AI harness as the software infrastructure that "wraps" around a LLM and enables it to act on tasks, not just respond to prompts. Here, Databricks provides a helpful analogy: where the AI model acts as the "brain" that generates reasoning and decisions, while the harness is everything around the model that helps the agent operate safely and reliably, including:

- **Tools:** APIs, code execution, search, databases and business applications
- **Memory:** Prior context, user preferences and workflow history
- **Workspace:** Files, data, environments and systems the agent can access
- **Guardrails:** Permissions, policies, approvals and monitoring

FIGURE 11. Higher-Level Agentic Harness Overview



Source: LangChain

Hence, agentic harnesses share similar principals to AI frameworks (e.g., augmenting AI models to address use cases) and sometimes these terms are used interchangeably, but are typically distinguished by the level of abstraction and user onus. AI frameworks operate at a lower level of abstraction, with low-level primitives (e.g., nodes, state) that require users to write the code and more granularly control application flow and chains. AI harnesses operate at a higher level

of abstraction and typically include the runtime environment (sandboxes, loops) that runs and manages an AI agent. Hence, users typically configure agentic harnesses as opposed to writing code, which takes complexity and onus away from the user. Put differently, harnesses sit one layer above AI frameworks and are closer to a configurable AI application as opposed to a framework that helps software developers build AI applications. Hence, we are seeing more proprietary agentic harness solutions that are monetizable, while AI frameworks tend to be more community-driven and open-source, where monetization is more implicit and moves up or down the stack. Examples of AI harnesses include OpenAI Agents SDK, Claude Code (Anthropic), Cursor (OpenAI), Pi, Cursor (SpaceX), Omnigent (Databricks/open-source) and CrewAI.

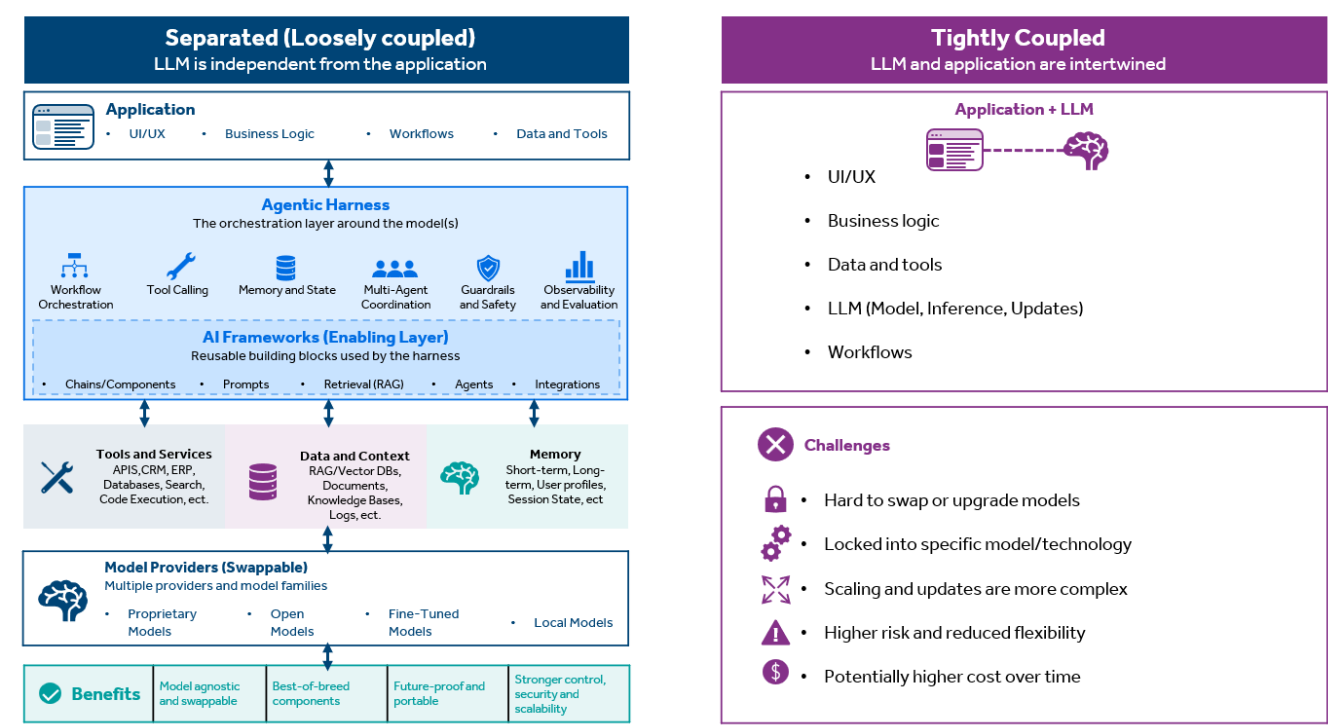
Why Are Agentic Harnesses Important?

The importance of agentic harnesses can be highlighted by one measurable idea: the same AI model with a different (better) harness yields different (better) results. For example, NVIDIA in a recent developer [blog](#) noted that harness design alone can account for double-digit swings in benchmark results and significant differences in token cost, with the same underlying model. Here, we emphasize that the "best" harness is not a complete thought as harness performance is use-case specific. For example, an agentic harness designed for coding will have better performance on coding tasks as opposed to a harness that caters to a specific vertical specific use case. For more information on this emerging layer, see [AI Frontiers: Introducing the AI Harness](#) (6/2/26).

Understanding the Paradigm Shift

The first generation of generative AI applications was relatively tightly coupled to individual model providers. An application might call a proprietary model provider's API directly, with prompts, tool definitions, structured outputs, error handling, and application behavior optimized around that provider's models. Hence, changing the underlying model could require meaningful application-level reengineering. The emerging architecture inserts several layers between the application and the model: application and workflow logic sits at the top; a model harness manages context, memory, retrieval, tools, prompts, and output validation; a gateway manages authentication, governance, budgets, and reliability; a router determines which model or provider should receive the request; and inference infrastructure ultimately executes the selected model.

FIGURE 12. Advancements in AI Frameworks and Harnesses Have Allowed the Separation Between Models and Applications



Source: Barclays Research

More recently, these frameworks have evolved alongside unified model APIs, AI gateways, inference engines, evaluation platforms, and model routers (discussed in later section) to create a broader model “separation” between the application and the underlying foundation model. This architecture allows developers to define application logic, context, tools, retrieval, governance, and evaluation independently of a specific model, while standardized interfaces translate those requirements across different proprietary and open-weight models. As a result, building sophisticated AI systems has become easier and more modular, lowering the cost of switching models and allowing enterprises and software vendors to select or dynamically route among models based on quality, cost, latency, privacy, or task requirements rather than designing the application around a single model provider.

Several recent technology developments have made this separation increasingly practical. Unified APIs and gateways from platforms such as OpenRouter, LiteLLM, Vercel, Cloudflare, and Portkey normalize many of the basic differences between model providers, allowing developers to access multiple models through a common interface. Serving engines such as vLLM and SGLang have simultaneously made open-weight models easier and more economical to operate by providing capabilities such as continuous batching, KV-cache management, prefix caching, quantization, parallelism, and OpenAI-compatible endpoints. Agent frameworks such as LangGraph, LlamaIndex, and Semantic Kernel increasingly place workflow state, retrieval, tools, and application logic above the model itself. standards such as Model Context Protocol further separate access to tools and enterprise context from any individual foundation model.

Challenges and Considerations

The architecture does not make models completely interchangeable. Basic API calls, authentication, streaming, and message formats are becoming increasingly standardized, but important behavioral differences remain. Models differ in reasoning behavior, tool use, structured-output reliability, multimodal capabilities, context handling, prompt sensitivity,

caching mechanics, safety policies, and proprietary features. A unified API can therefore abstract much of the infrastructure surrounding a model without fully abstracting the model's behavior. For sophisticated applications, developers are likely to retain model-specific optimizations and provider-native functionality where those capabilities materially improve application performance.

Implications

We believe this represents an important architectural shift in the AI stack. Rather than embedding the logic of a specific model directly into an application, developers can increasingly build against an abstraction layer, or “harness,” that sits between the application and the underlying models. This layer can translate application requirements into model calls and determine which model or inference provider should execute a given task. Over time, this architecture could reduce model switching costs, make open-weight models substantially easier to deploy, and allow enterprises and software vendors to retain greater control over their intelligence stacks.

This surrounding infrastructure is particularly important for open-weight models because model weights alone do not constitute a production AI service. Downloading an open model does not inherently provide autoscaling, batching, authentication, rate limiting, retries, structured-output enforcement, tool integration, safety controls, observability, evaluation, enterprise governance, or production reliability. In effect, the emerging model harness supplies many of the capabilities that proprietary model providers historically bundled into their APIs. Improvements in inference engines, gateways, orchestration frameworks, and evaluation tooling therefore make open models more viable substitutes for proprietary APIs even if the underlying models themselves have not become fully interchangeable.

The strategically important control point may consequently migrate upward in the stack. The application or harness increasingly knows what the user is trying to accomplish, what enterprise data and tools are available, which models are permitted, how different models have historically performed on the workload, and what constitutes a successful business outcome. That information can be more valuable for routing than generic benchmark performance. The application owner can use it to continuously evaluate models and substitute new models as their relative quality, cost, and latency change.

We therefore view the emerging model framework and harness layer less as a single new software category and more as an architectural control layer spanning orchestration, gateways, routing, evaluation, and inference management. The most durable economic value may accrue not to a standalone router, but to whichever platform controls application context, workflow state, evaluation data, governance, and the model-selection decision. If this architecture continues to mature, enterprises and software vendors should gain greater control over their intelligence stacks, open models should become easier to incorporate into production workloads, and foundation-model providers may increasingly compete to supply intelligence into applications whose architecture and customer relationship are controlled elsewhere.

Model Selection: Open vs. Closed Models

When comparing open and closed models, there are trade-offs to consider ([Figure 13](#)). With open-weight models, the trained parameters (weights) are published for users to download (often for free, although licensing varies). This enables greater flexibility in deployment, including customization through fine-tuning.

Self-hosting open-weight models involves owning GPUs or leasing them from a cloud provider and running inference through a user's own serving stack (i.e., the software layer used to serve model requests). Importantly, self-hosting can provide more control over data handling, since

the data is not sent to an external model provider. Small models can also be deployed on edge devices, which can reduce latency. However, self-hosting also introduces complexity, as organizations must have the technical expertise and resources to deploy, monitor, and maintain the infrastructure.

To reduce this operational burden, managed hosting solutions allow organizations to use open-weight models without the responsibility for the underlying infrastructure. Meanwhile, closed models can offer simpler access to frontier capabilities. Leading frontier models continue to outperform open-weight alternatives on some of the most complex tasks, such as advanced reasoning and agentic workflows.

The cost structures of closed models and self-hosted open-weight models differ, with the cost of using a closed model determined by price per token x token consumption. With self-hosted open-weight models, the model itself is often free to download, but users pay for the cost of compute and operations. This can become significantly more cost-effective when GPU utilization is high enough to justify the infrastructure cost. We are also seeing proprietary model labs introducing new optimizations into their AI stacks to drive better price performance and these labs also lowering prices for some of their models. In the long-run, cost per task will increasingly become the most important factor for model adoption as basic capabilities become more monetized.

While much of the initial debate has been on open vs. closed models, in our view, many organizations are likely to adopt hybrid approaches. With [model routing](#), users can match workloads to the most appropriate model based on performance requirements, cost, latency, and security considerations.

FIGURE 13. Open vs. Closed Model Comparison

	Open-Weight Models	Closed Models	Key Takeaway
Performance	Performance gap has narrowed significantly, with leading open models competitive on many tasks	Generally lead on frontier capabilities, particularly advanced reasoning, multimodal capabilities, and agentic workflows	Closed models lead at the frontier, but the gap has narrowed
Cost	Most models are free to download, but self-hosting incurs infrastructure, compute, and operating costs	Typically subscription-based, usage-based, or a combination of both	Open models can offer cost advantages at higher inference volumes
Ease of Deployment	Self-hosting requires expertise to deploy and maintain, although managed hosting can simplify deployment	Easily accessible through APIs or managed platforms	Closed models are typically easier and faster to deploy
Data Protection	Can be self-hosted, providing greater control over data security, privacy, and compliance requirements	Data is processed within the provider's environment, although enterprise deployments often offer robust protections	Open models may be preferred in highly regulated industries or when data sovereignty is a priority
Customization	Can be fine-tuned and adapted using proprietary data; greater control over model behavior	More limited customization, typically through managed fine-tuning or provider-supported tools	Open models can be better suited for highly specialized or domain-specific applications
Model Updates & Vendor Lock-In	Organizations control when to adopt new models and updates, but are responsible for implementation and maintenance	Continuous access to provider's latest models and improvements, but subject to vendor decisions and potential vendor lock-in	Open models offer more control; closed models offer simpler access to the latest capabilities
Latency & Reliability	Organizations have greater control over latency and reliability, but are responsible for managing infrastructure; can run small models on edge devices	Dependent on the provider's infrastructure, but leading providers typically offer highly optimized, highly reliable deployments	Open models offer greater control; closed models offer managed reliability and scale

Source: Barclays Research

Insights from Companies Leveraging Open-Weight Models

Several companies have discussed the advantages of using open-weight models, highlighting opportunities to adopt hybrid strategies and build proprietary, domain-specific models.

- **Thomson Reuters** recently launched its proprietary LLM, "Thomson," with the company investing \$40m to build on top of an open-source foundation model. Thomson is trained using the company's domain expertise and content from Westlaw, Practical Law, Checkpoint, and Reuters. According to the [press release](#), Thomson is already competitive with leading frontier models across a range of tasks while operating at a fraction of the cost. Notably, the model was trained on less than 10% of Thomson Reuters content so far, highlighting how domain specialization and proprietary datasets can support strong performance without relying solely on larger models.
- **Pinterest's** approach to model deployment includes compact, fit-for-purpose proprietary models, open-source models, and in some select cases, closed models. With open models, Pinterest is achieving cost per transaction at less than 8% of the cost of comparable closed models. Open-source models also outperform on company-specific use cases, benefiting from post-training on company data. Pinterest's routing layer optimizes model selection, with plans to expand this infrastructure to more internal use cases in 2H'26.
- **Coinbase** reduced AI spend by nearly 50% through a combination of model selection, routing, and caching. The company has started defaulting tasks to open-weight models like Z.AI's GLM 5.2 and Moonshot AI's Kimi K2.7, while encouraging engineers to choose the right model for the task. Coinbase also improved its cache hit rate from 5% to 60% and is making an effort to reduce token waste by keeping context lean and improving visibility into usage. These efforts have allowed the company to reduce AI spend even as token consumption rises.
- **Duolingo** is adopting open-source models where performance is comparable, enabling lower costs and broader deployment of AI features. For example, during its Q2'26 earnings call, the company highlighted open-source models as a key driver of lower video call costs. With the cost per video call now under \$0.01, Duolingo has been able to offer video call capabilities to more subscribers.
- **Hims & Hers** has highlighted the company's datasets as an advantage in its AI strategy. During an interview with CNBC, CEO Andrew Dudum discussed the opportunity for companies with resources and scale to combine open-weight models with proprietary datasets, highlighting the potential for cost savings upwards of 70-80% and better performance on company-specific use cases. Closed-loop datasets, such as following a patient through their healthcare journey, can serve as a valuable asset, enabling models to learn and improve over time.

Model Routing: Right Tool for the Right Job

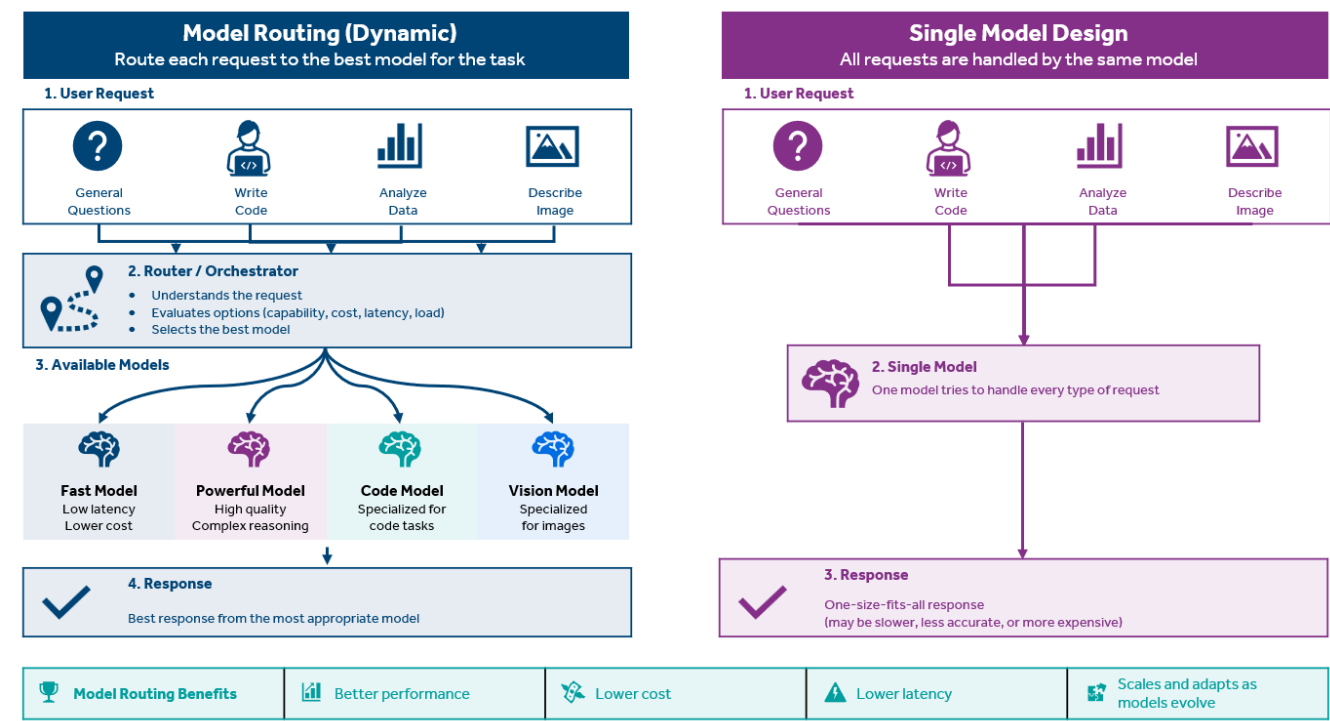
As new model releases continue to raise the bar on model performance and capabilities, there is also growing momentum around the idea that not all tasks need the most advanced (and expensive) models. Model routing services have gained popularity as enterprises begin managing AI spend more efficiently through software advancements.

What Is Model Routing in AI?

At its core, model routing (also known as LLM routing or inference routing) is the process of directing user queries or requests (such as a question to an AI chatbot) to different models based on some goal (performance, cost, latency) as opposed to relying on a single model for all tasks. In more modern implementations, this routing is done dynamically and considers the trade-off between performance, cost, and latency, which can be adjusted to the end-users

preferences between these trade-offs. The idea is that simple queries (e.g., what is the capital of New York) should be routed to smaller, cheaper models, while more complex queries and reasoning should be performed by more powerful models. Specific, domain tasks could also be sent to smaller, more specialized models (e.g., the small language model, or SLM trend), which could be fine-tuned open-weight/source models.

FIGURE 14. Higher-Level Model Routing Design Pattern vs. Single Model Design



Source: Barclays Research

Benefits of Model Routing Systems

The promises of model routing systems are lower inference costs, reduced LLM lock-in and better performance and latency. In addition to "choosing the cheapest/most efficient model for the job," models are often designed for different purposes (coding, text, image generation) and are different sizes. Hence, when building an AI application that has different purposes, model routing capabilities are becoming a key capability to most efficiently serve diverse queries at scale.

As an example of the benefits of model routing systems, an IBM [RedHat blog](#) highlights that benchmarks used by vLLM Semantic Router (described within) show that, with auto reasoning mode adjustment, using MMLU-Pro and Qwen3 30B model, there was a ~10% improvement in accuracy, ~47% reduction in latency, and ~49% reduction in token usage.

Dynamic model routing extends this abstraction further by turning model selection from a design-time architecture decision into a runtime optimization problem. Rather than choosing one model when an application is built, an application can maintain a pool of eligible models and select among them for each request, or potentially for each step within an agent workflow. For example, a simple extraction task might be sent to a small open model, routine summarization to an inexpensive proprietary model, coding to a specialist model, and difficult reasoning to a frontier model. Sensitive workloads could be routed to self-hosted models, international requests to approved regional endpoints, and traffic could automatically fail over to another provider during an outage.

Characteristics of Modern Model Routing Systems

In the context of LLMs, our research points to early model routing capabilities emerging in 2022-2023 that manually coded primitive routing flows (e.g., if token count > 100, send to X model, if <100 tokens, send to Y model). Since then (approx. 2025-2026), model routing capabilities have advanced, with today's systems evaluating many factors simultaneously, such as intent, complexity, safety, context length, latency, cost budget, and historical success, to dynamically route requests to the most suitable model. For a closer look at routing techniques, we point to [AI Inference Trending Towards "The Right Tool for the Right Job" Mindset](#), and note that modern systems often combine multiple approaches.

Implications

The direction of travel has important competitive implications as the economic incentive can be meaningful because model pricing varies substantially. As the abstraction layer improves, the cost of switching between models should decline and enterprises can increasingly treat models as alternative suppliers of intelligence rather than fixed dependencies. Open models can provide negotiating leverage against proprietary providers, while routing allows applications to consume frontier intelligence selectively rather than paying frontier-model economics for every request. This creates the potential for a barbell architecture in which inexpensive open or smaller models absorb high-volume routine inference while expensive frontier models are reserved for tasks where their incremental intelligence creates sufficient economic value. The potential value of routing therefore comes primarily from optimizing model selection rather than from the routing software itself. This does not necessarily commoditize foundation models, particularly where leading models retain meaningful performance advantages, but it can weaken the relationship between model leadership and ownership of the end customer. However, as previously discussed, leading model providers are creating a portfolio of models with different performance/cost trade-offs, which helps insulate them from this dynamic and ultimately leads to benefits to the end consumer from a cost per task perspective.

Examples of Model Routing Services

In [Figure 15](#), we detail examples of model routing services from industry vendors and the open-source community. We note that this list is not comprehensive, particularly for the open-source technologies, as there has been a rise in open-source and GitHub repositories related to LLM routing capabilities. Large model providers like OpenAI and Anthropic have model routing capabilities, but those have been centric to their own portfolio of proprietary models. On the AI compute side, we note DigitalOcean's new Inference Engine as an interesting example that highlights how the compute layer is taking advantage of model routing capabilities (see [DigitalOcean: Inference Engine Launch Headlines DOCN's AI-Focused Conference Keynote, 4/29/26](#)).

FIGURE 15. Examples of model routing services

Proprietary/Enterprise

DigitalOcean Inference Engine	DigitalOcean released its Inference Engine solution during its Deploy Conference in April 2026, which includes Inference Router (in public preview). Inference Router allows customers to route serverless inference and dedicated inference requests to foundation models using configurable rules to optimize model costs and latency.
IBM	IBM offers watsonx Model Gateway that offers a unified, OpenAI-compatible API endpoint that dynamically routes inference requests across multiple model providers and has an inference router built into its AI Optimizer for Z and LinuxOne that leverages tags and other specified metadata to intelligently route queries to the suitable model.
Microsoft Foundry Model Router	Microsoft offers Microsoft Foundry Model Router that uses an optimized AI model (router itself) to analyze system messages, history and complexity in seconds to send incoming prompts to the best fit model. The service offers different modes (Balance, Cost or Quality) and automatic failover to the next best model if issues occur. Microsoft documentation discusses this can be embedded in existing applications using the Azure AI Foundry portal or the OpenAI Python SDK.
Amazon Bedrock	Through its Amazon Bedrock offering, Amazon features Intelligent Prompt Routing, which provides a single serverless endpoint that automatically routes requests between different foundational models within the same family, but notably does not route between open-source models and different model providers. We do note the idea that users can leverage open-source router technologies via APIs, etc. that can run on Amazon's and other providers cloud infrastructure.
Google	Google has built-in LLM routing within its consumer and enterprise services such as Gemini Advanced and also offers Google Kubernetes Engine Inference Gateway that allows users to build intelligent routing solutions.
OpenRouter	A large cloud-managed cloud routing marketplace that connects to hundreds of models and automatically routes queries based on speed, cost and availability, also featuring consolidated billing.
Vercel	Vecel's AI Gateway solution provides an API to access hundreds of models, static and dynamic routing capabilities and failover capabilities.
Portkey	Portkey markets itself as a generative AI platform and gateway used by enterprises to build, secure and monitor AI applications. It offers an AI gateway with conditional and task based routing capabilities and automated fallbacks. The company has also open-sourced parts of its core gateway stack.
Braintrust Gateway	Braintrust is primarily an evaluation and observability platform that also has an integrated gateway that supports multi-provider access, caching and logging.

Open-Source

LiteLLM (Primary Language Python)	An open-source proxy service driven by the core team at BerriAI that provides an OpenAI-compatible interface across over 100 LLMs in addition to load balancing, fallbacks, retries and caching.
vLLM Semantic Router	An open-source decision making and routing layer for AI agents and LLMs that uses classifiers like ModernBERT to measure query intent and complexity to route tasks appropriately. The project is highly collaborative, driven by over 100 global contributors alongside organizations including Red Hat, IBM Research, AMD and Hugging Face.
Bifrost (Primary Language Go)	Open-source routing gateway provided from Maxim AI that provides unified access to 20+ providers through a unified API that allows users to configure routing rules on a virtual key and create automatic fallbacks.

Source: Company Documents, Barclays Research

AI Strategies: Focus on ROI

In addition to decisions around how AI is deployed from a technical perspective, companies are also reconsidering AI applications. The initial phase of enterprise adoption encouraged employees to experiment, prioritizing access and usage. This led to token consumption sometimes treated as a proxy for productivity, driving spend higher. In recent months, this approach has evolved, with companies moving away from the tokenmaxxing trend to focus on controlling costs through more targeted deployment strategies.

The Next Phase of AI Adoption: From Experimentation to Strategic Deployment

Some companies have responded to elevated spend by introducing restrictions on employee AI usage. While these announcements have gained attention, in our view, the relevant trend is not just the shift from tokenmaxxing to restrictions. Instead, we see the next stage of AI adoption centering on more targeted implementation. For example, Uber spent its 2026 AI budget in just four months, leading the company to introduce [\\$1,500 monthly spending limits](#) per AI coding tool. Importantly, however, Uber's approach extends beyond cost controls. The company has introduced [agentic pods](#), which pair AI-proficient engineers with domain experts in business areas like finance, legal, and marketing to uncover opportunities for automation and workflow redesign.

Although employee restrictions can help control spend, in our view, long-term success will depend on deploying AI more effectively rather than just constraining deployment. For example, Coinbase has discussed improving visibility into token usage without introducing constraints, encouraging usage where it leads to impact. We expect this broader focus on execution to ultimately influence which companies are capturing the value, elevating the importance of factors such as proprietary data, technology readiness, and AI talent, as we [detail below](#).

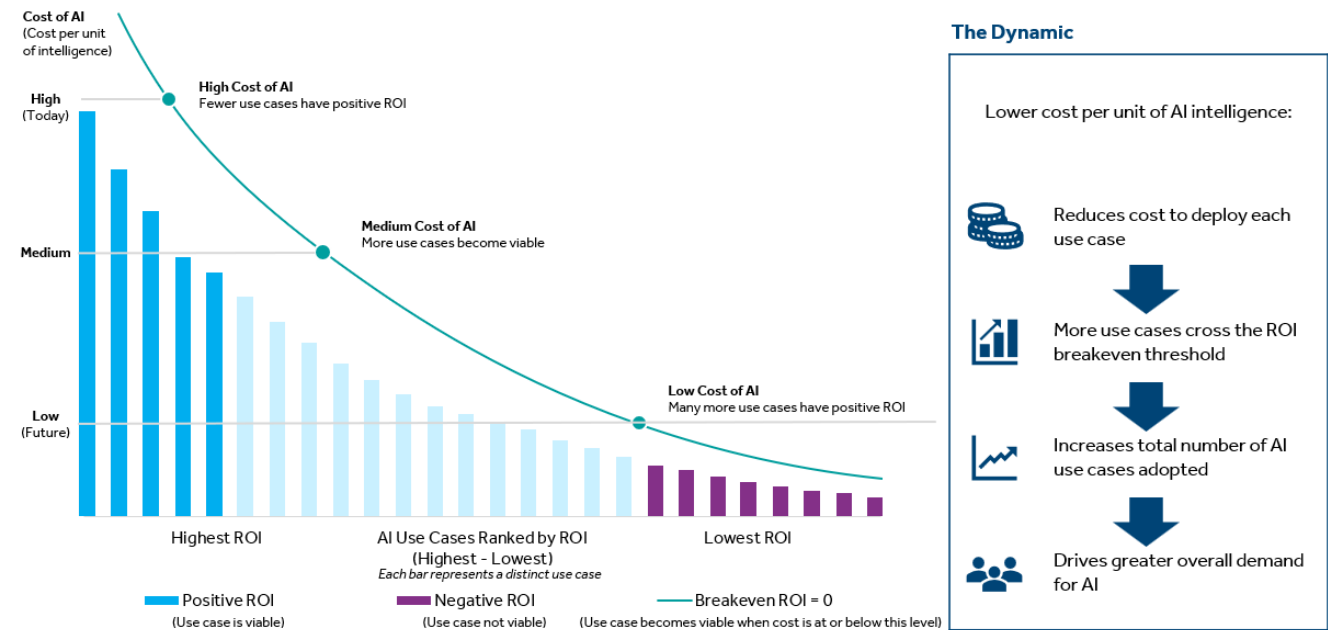
ROI Discipline Does Not Equal Less Demand

Importantly, managing AI spend does not necessarily mean reducing it. Although much of this spending is drawn against relatively fixed budget pools, as AI unlocks new revenue opportunities and cost savings, companies may reinvest in further deployment. Lower AI costs also expand the number of use cases with a positive ROI. In turn, Jevons Paradox should support higher demand even in an era focused on strategic AI deployment and ROI ([Figure 16](#)).

FIGURE 16. Visualizing the Jevons Paradox in AI adoption

Jevons Paradox in AI: Lower Cost Drive Exponentially Higher Demand

As the cost of AI falls, the number of use cases with positive POL expands substantially, driving greater overall demand for AI.



Source: Barclays Research

INVESTOR GUIDEBOOK: WHO CAPTURES THE VALUE?

Enterprises Executing on AI Adoption

As the focus shifts from access and experimentation toward measurable business outcomes, companies are placing greater emphasis on demonstrating return on investment. For investors, this raises important questions about both the timing of AI impacts and which companies are best positioned to capture the resulting value. While stricter ROI requirements may create some near-term friction, this shift also increases the importance of factors such as proprietary data, technical talent, and organizational culture in determining which organizations execute most effectively and realize lasting competitive advantages.

Timing of AI Value Realization

In our view, a key inflection point for the theme is when AI benefits become more visibly tied to core financial metrics. Recent [transcript analysis](#) from our U.S. Equity Strategy team reveals that a wide gap between AI ambition and execution exists today. Although nearly half of S&P 500 companies talked about AI during Q2'26 earnings, only 12% quantified AI-related value capture.

In the near term, it's possible that more restricted usage and an increased focus on ROI could slow the pace of deployment or raise the threshold for what management teams are willing to disclose. As companies apply greater scrutiny, some may be less willing to quantify expected cost savings or revenue opportunities and as a result, evidence of AI value creation could take longer to emerge in company disclosures, reported financial results, or management commentary, even if underlying deployment activity and benefits continue to improve. Over time, however, we view this shift as a sign of maturation. A more disciplined approach to AI deployment should encourage companies to identify where AI creates the greatest value and concentrate investment on the highest-return use cases.

Execution Becomes the Differentiator

As the focus shifts from AI adoption to value realization, this also raises the question of which companies are best positioned to capture the benefits. A key debate here is whether AI will act as a force multiplier or leveler. Our work on AI Disruption suggests scale can influence company positioning, with industry metrics indicating larger companies are adopting AI at higher rates. According to the Ramp AI Index, a greater share of large companies are spending on AI, with 66.8% of large companies paying for AI models, subscriptions, tokens, and APIs vs. 50.1% of small businesses, as of August 2026. Data from the Business Trends and Outlook Survey (BTOS) and Survey of Business Uncertainty (SBU) similarly show that the largest firms have the highest adoption rates. However, adoption metrics alone do not resolve whether AI advantages will

“From the adoption metrics alone, it is hard to say whether AI-related gains are accruing ly to the largest firms or whether AI is serving as an equalizer of sorts and helping smaller enterprises compete more effectively.”

— FEDS Notes – Monitoring AI Adoption in the US Economy – Jeffrey S. Allen (April 03, 2026)

In our view, the more important question is not simply which companies are adopting AI, but which companies can translate adoption into measurable business outcomes. As the availability

of highly-capable, low cost models expands, model access alone is becoming less of a source of differentiation. However, the ability to deploy and scale AI effectively is likely to remain unevenly distributed across organizations.

As a result, the next phase of AI value realization will depend more on execution. Key differentiators will include factors like proprietary data, technology readiness, and AI talent. Companies that can combine these capabilities may be better positioned to capture economic value regardless of which models they rely on.

Company Landscape: 70 Companies Adopting AI

With this execution-focused lens, we build on our *Positioning for AI Disruption* work to identify 70 companies across various stages of AI adoption ([Figure 17](#)). We provide details on implementation progress, targets, and company-specific factors that could help support successful execution. We focus on sectors outside of the Tech industry; given high levels of AI adoption within Tech sectors, our company landscape assesses the opportunity for competitive differentiation across Industrials, Energy, Financials, Consumer, and Healthcare. For Tech insights as well as further company-level detail, we point to the [U.S.](#) and [EU](#) editions of our deep dive, which include comprehensive industry profiles.

FIGURE 17. Company Landscape: 70 Companies Adopting AI

Sector	Company	Industry	Barclays Analyst	Stage of AI Adoption	AI Implementation: Progress & Targets
Industrials & Materials	Thomson Reuters	U.S. Business & Professional Services	Manav Patnaik	Proven Execution	AI is enhancing Thomson Reuters' core Legal, Tax, and Corporates businesses while creating a new opportunity through CoCounsel. AI has already transformed the organic growth profile of the three major segments, which is moving toward approximately 10%, demonstrating the company's ability to monetize AI-enabled efficiency gains in mission-critical professional workflows.
	C.H. Robinson	North America Airlines & Transportation	Brandon Oglenski	Proven Execution	C.H. Robinson is leveraging technological improvements in carrier and revenue management to disrupt freight brokerage and make it increasingly difficult for smaller brokers to compete. AI-enabled lean principles are already improving net revenue and operating margins versus peers.
	GXO Logistics	North America Airlines & Transportation	Brandon Oglenski	Scaling	GXO already uses robots in automated warehousing applications to sort, deliver, and process customer inventories. The company plans to develop physical AI applications that expand robotics capabilities and increase labor productivity.
	Carvana	U.S. Auto Retail	John Babcock	Proven Execution	Carvana is an online retailer of used vehicles. It is increasingly using AI to help drive costs lower. As an example, the company has used AI to develop software that helped it streamline operations and reduce bottlenecks at its inspection & reconditioning centers (IRCs). CVNA has also discussed using AI to help with centralized customer care, transaction processing, and vehicle title and registration, among other areas.
	OPENLANE	U.S. Auto Retail	John Babcock	Scaling	OPENLANE operates a leading digital wholesale auction platform and is using AI to develop features that help buyers and sellers assess vehicle value and condition. The company is also working to improve transaction speed and efficiency to make digital auctions more seamless and capture share from physical auctions.
	ACV Auctions	U.S. Auto Retail	John Babcock	Scaling	ACV is rolling out Project Viper, which uses AI-driven inspection technology to automate vehicle condition reporting and provide more detailed vehicle-condition data. Project Viper will give dealers more transparency in the vehicle acquisition process, and potentially help dealers identify opportunities to upsell in Parts & Service. ACVA is already leveraging Project Viper for vehicle inspections at its remarketing centers and plans to roll out over 500 of the units to dealers next year. Its value-added services also use transaction-level data to improve the buyer experience and marketplace returns.
	Republic Services	North America Environmental Services	William Grippin	Scaling	Republic Services is deploying AI through its RISE digital dispatch platform, algorithm-driven fleet routing, virtual customer-service agents, and smart pricing. These initiatives are supporting productivity, reducing call-center workloads, and contributing to a potential long-term cost-savings opportunity of more than \$100m.
	ACS	European Construction, Building Materials & Infrastructure	Katherine Hearne	Scaling	ACS is deploying AI internally to improve project transparency (real-time tracking platforms) and streamline workflows via custom GPTs.
	Vinci	European Construction, Building Materials & Infrastructure	Pierre Rousseau	Scaling	Vinci has a broad and mature internal AI program, spanning tender selection, cost estimation, contract analysis (LLMs), scheduling, safety monitoring, and predictive maintenance. The company has framed AI as a tool to defend margins and reduce project risk variability across its large and diversified orderbook.
	Verisure	European Business Services	James Rowland Clark	Scaling	Verisure is using AI to improve the performance of its customer contact centres and integrating AI into its cameras and sensors to improve verification accuracy. These capabilities can also reduce monitoring agents' workloads and the number of physical call-outs required. In the near term, Verisure's products should be more advanced than those of its peers.
	Rentokil	European Business Services	James Rose	Early Stage	Rentokil can use AI to improve marketing and sales lead monitoring, customer-facing processes such as scheduling and billing, and back-office functions.
	Experian	European Business Services	James Rose	Scaling	Experian can use AI to accelerate its product development pipeline and improve the effectiveness of existing analytical platforms.
	DSV	European Transportation	Marco Limite	Scaling	DSV has been a first mover, with investment in an AI factory beginning several years ago and early deployment in customs, booking, and quoting already in progress.
	Airbus	European Aerospace & Defence	Milene Kerner	Scaling	AI is already deployed in production planning, quality inspection, and supply-chain resilience, notably on A320 and A350 programmes. The benefit is primarily lower execution risk and improved delivery credibility during the current production ramp-up.
	Safran	European Aerospace & Defence	Milene Kerner	Scaling	Safran's large installed base, proprietary flight-hour data, and increasingly digitalised MRO platforms enable AI-driven predictive maintenance, shop-visit optimisation, and spare-parts forecasting. These applications support aftermarket margins, availability, and working capital.

Sector	Company	Industry	Barclays Analyst	Stage of AI Adoption	AI Implementation: Progress & Targets
Energy & Utilities	SLB	U.S. Energy Services & Technology	J. David Anderson	Scaling	SLB has moved customer subsurface models and well data to the cloud and created the Delphi operating system to connect that data with applications. Lumi integrates generative AI with energy workflows, while Tela uses LLMs and domain foundation models to automate processes and adapt workflows in real time.
	bp	European Integrated Energy	Lydia Rainforth	Proven Execution	bp is supported by more than a decade of digitalisation and a strategic partnership with Palantir in the upstream division. This foundation is already translating into measurable operational gains, with bp increasing operated production by around 4% between 2022 and 2024 and preventing roughly 10% more downtime through enhanced surveillance, real-time monitoring, and analysis. bp targets having 80% of its platforms unmanned by 2030, and our European Integrated Energy team estimates the company could unlock up to \$10bn in productivity gains by 2030.
	TotalEnergies	European Integrated Energy	Lydia Rainforth	Scaling	Data is a core asset for TotalEnergies, which benefits from a long history of data accumulation and generates very large volumes daily. The company has invested heavily over the past decade in its data infrastructure, and with data foundations now significantly improved, the focus is shifting towards scaling deployment. The company has a planned \$1bn investment in AI over 2026–2028, targeting improvements in production, sales, and margins.
	Tecnicas Reunidas	European Energy Services	Mick Pickup	Early Stage	Tecnicas Reunidas estimates that approximately 60% of internal man-hours across engineering, procurement, construction, site operations, and project control can be automated with current robotics and AI, creating the potential for EUR200m per year of cost efficiencies over time.
	Viridien	European Energy Services	Mick Pickup	Scaling	Viridien embeds AI within physics-based imaging workflows to improve noise removal, QC, and cycle-time efficiency. The company's Geoscience unit remains structurally differentiated, driven by talent and compute capacity, with AI seen as an enhancer rather than a replacement.
	Maire	European Energy Services	Mick Pickup	Proven Execution	More than 7,000 engineers use AI tools, redirecting roughly 1m man-hours in 2025 to higher-value tasks. AI is being applied to bid estimation, query resolution, forecasting, vendor intelligence, and cost accuracy, while Next Brains, a 200-expert AI hub, will support autonomous plant operations.
Financials	Bank of America	U.S. Large-Cap Banks	Jason Goldberg	Proven Execution	Erica, Bank of America's client-facing virtual assistant, serves 20m active clients and has facilitated more than 2.5bn interactions since inception. Internal adoption exceeds 90%, routine IT service-desk calls have declined by more than 50%, and software developers are achieving efficiency gains through AI coding and optimization tools.
	JPMorgan Chase	U.S. Large-Cap Banks	Jason Goldberg	Proven Execution	Approximately 150,000 employees use AI tools daily, with management estimating up to \$2bn of annual cost savings. Early AI implementations have reduced manual KYC and AML operations by roughly 40%, while agentic systems continuously monitor client profiles and triage high-risk anomalies.
	First Horizon	U.S. Mid-Cap Banks	Jared Shaw	Proven Execution	First Horizon is already finding both efficiency- and growth-focused AI use cases. Management has discussed AI benefits, including better efficiency in software development for its new consumer app and improvements in customer prospecting, noting that the win rates in AI-driven targeting have been strong in generating incremental relationships.
	Associated Banc-Corp	U.S. Mid-Cap Banks	Jared Shaw	Early Stage	Leadership has communicated urgency around AI adoption throughout the organization. Several strategic technology upgrades have been implemented since 2021, and the culture of innovation in product development has advanced, creating a stronger foundation for AI-enabled growth in new markets.
	Royal Bank of Canada	Canada & Latin America Banks	Brian Morton	Proven Execution	Royal Bank of Canada has established a target of C\$700m to C\$1bn of enterprise value generated through the implementation of AI. The company has noted data and platform scale, its AI platform, and differentiated insights as foundations for this target.
	TD Bank	Canada & Latin America Banks	Brian Morton	Proven Execution	TD has established a C\$1bn target for value from AI, split between roughly C\$500m of revenue and C\$500m of cost savings. Use cases include client experience and personalization, financial-crime prevention, client-facing assistants, and residential-lending and wealth operations.
	Banorte	Canada & Latin America Banks	Brian Morton	Early Stage	Sizable digital and technology investments provide the foundation for further AI development and create a competitive advantage versus domestic peers.
	Arthur J Gallagher	North America Insurance Life & Non-Life	Alex Scott	Scaling	Arthur J. Gallagher is rolling out AI initiatives across the organization to drive efficiency and improve the quality and responsiveness of client service. Its history of margin improvement and willingness to shift roles offshore support the realization of AI-driven productivity gains.
	The Hartford Group	North America Insurance Life & Non-Life	Alex Scott	Scaling	Hartford has one of the most scaled platforms in small commercial insurance and has invested more heavily in technology than many peers. Generative-search optimization and automation across high-volume submissions should improve customer acquisition, quote flow, and underwriting efficiency.

Sector	Company	Industry	Barclays Analyst	Stage of AI Adoption	AI Implementation: Progress & Targets
	Rocket Companies	U.S. Consumer Finance	Terry Ma	Proven Execution	Rocket's broad deployment of automation includes AI-enabled loan officers, automated document review, and streamlined credit approvals. These tools have enabled loan officers to serve 40% more clients, improved conversion by double-digits compared to one-year ago and contributed to roughly \$2bn in incremental volume per month.
	Robinhood	U.S. Brokers, Asset Managers & Exchanges	Benjamin Budish	Scaling	Robinhood is rolling out Cortex, an AI-powered assistant and analytics suite that provides portfolio digests for market analysis and helps users convert ideas into executable trades. AI is also expected to support more personalized customer offers and wallet-share gains.
	Santander	European Banks	Cecilia Romero Reyes	Proven Execution	Santander combines scale with a modernising multi-market technology platform that allows AI investments to be deployed once and reused across the group. It links roughly EUR700m of cost savings and EUR300m of incremental revenue to AI within its efficiency plan.
	ING	European Banks	Namita Samtani	Proven Execution	ING benefits from a centralised cloud-based data and analytics platform that allows it to scale AI across the organisation more effectively than peers. Management frames AI around three pillars: equipping its c.60,000 employees with AI capabilities, improving the efficiency of manual and operationally intensive processes such as AML/KYC, where FTE reductions have already been observed, and redesigning end-to-end processes rather than simply layering AI onto existing workflows. The bank is already seeing tangible benefits from these efforts, with agentic AI deployed in Dutch mortgages helping accelerate approval times and improve processing capacity. Together with its scalable technology platform, ING continues to target c.€350m of incremental cost efficiencies in 2026 from digitalisation, operational efficiencies, and AI adoption.
	LSEG	European Diversified Financials	Michael Sanderson	Scaling	LSEG collects, manages, and distributes a substantial volume of data and is using AI to enhance its client proposition and its own data-management processes. Increasing demand from AI agents and tools for curated data is creating new distribution channels, customer opportunities, and add-on sales.
	Man Group	European Diversified Financials	Michael Sanderson	Proven Execution	A core differentiator for Man Group has been investment in talent and technology, with structured data management powering its quantitative strategies. The company is dedicating resources to AI to enhance research, deliver scalability, and increase automation, building on more than 100 proprietary plug-ins and a partnership with Anthropic.
	AXA	European Insurance	Claudia Gaspari	Scaling	AXA combines broad AI adoption with increasing deployment at scale. The Group is scaling more than 400 AI use cases, including 60+ agentic AI applications across underwriting, contact centres and claims, supported by partnerships with Microsoft and Publicis Sapient. AXA's strategy targets €500m-€700m of recurring annual pre-tax AI value by 2029, supporting lower costs, better fraud detection, and higher productivity.
	Allianz	European Insurance	Claudia Gaspari	Scaling	Allianz combines scale with deep AI adoption. AllianzGPT serves 60,000+ active users, while its partnership with Anthropic extends AI across employee workflows, software development, and agentic process automation. With more than 900 AI use cases registered globally and one of the industry's deepest AI talent pools, Allianz appears well positioned to translate AI adoption into productivity and growth.
	Unibail-Rodamco-Westfield	European Real Estate	Paul May	Proven Execution	Unibail has already started using AI to increase revenue. Westfield Rise, the in-house media and experiential division of URW, launched a programmatic digital-out-of-home (DOOH) platform in June 2025 and uses anonymised AI video analysis to accurately target shoppers based on age, gender, and behavioural cues.
Consumer	The Coca-Cola Company	U.S. Consumer Staples	Lauren Lieberman	Proven Execution	Roughly two-thirds of Coca-Cola system customers have shifted to digital ordering, supporting extensive data capture and predictive analytics. Suggested orders have driven a 4-6% sales uplift, while North America pilots of dynamic media targeting have increased media effectiveness by 15-50%.
	Home Depot	U.S. Broadlines, Hardlines & Food Retail	Seth Sigman	Scaling	Home Depot is leveraging tech and AI to drive more real time decision-making and greater productivity. HD has discussed increased consumer and associate usage of its Magic Apron AI assistant and is piloting a new AI tool with enhanced features and in-store functionality.
	Walmart Inc.	U.S. Broadlines, Hardlines & Food Retail	Seth Sigman	Scaling	Walmart is structurally advantaged by its omnichannel scale and the large amount of data available to develop sophisticated AI tools for the customer experience. AI is also supporting automation across stores and the supply chain, as well as the expansion of higher-margin advertising, membership, and fulfillment businesses.
	Wayfair Inc.	U.S. Broadlines, Hardlines & Food Retail	Seth Sigman	Scaling	Wayfair is deploying AI across an extensive catalog on its website/mobile app, its proprietary Castlegate logistics network, and engineering teams to drive conversion, customer loyalty, labor efficiency, and ad optimization.

Sector	Company	Industry	Barclays Analyst	Stage of AI Adoption	AI Implementation: Progress & Targets
	Ralph Lauren	U.S. Specialty Retail, Apparel & Footwear	Adrienne Yih	Scaling	Ralph Lauren is well positioned to capture near-term benefits through higher conversion, improved full-price selling, and richer first-party data. The company is deploying AI across both consumer-facing and back-end use cases, including predictive buying and Ask Ralph, its generative AI shopping tool, which is already driving meaningful engagement.
	Tapestry	U.S. Specialty Retail, Apparel & Footwear	Adrienne Yih	Scaling	Tapestry began making investments to strengthen its capabilities in customer analytics, inventory management, and supply chain management well before AI became the dominant theme it is today. Tapestry's mature, data-rich technology stack can help the company scale AI faster across established enterprise workflows.
	Ulta Beauty	U.S. Specialty Retail, Apparel & Footwear	Adrienne Yih	Scaling	Ulta is using AI across conversational guest service, marketing personalization, order management, and inventory optimization. These applications support service efficiency, targeting, conversion, loyalty engagement, and lower out-of-stock and markdown risk.
	Royal Caribbean Group	U.S. Gaming, Leisure & Lodging	Brandt Montour	Scaling	Royal Caribbean has been at the forefront of modernizing its technology stack, positioning the company to integrate AI more quickly across distribution, trip planning, and customer engagement. Industry-leading data and digital capabilities also support direct collaboration with AI travel platforms.
	Arca Continental	Americas Agribusiness & Mexico Consumer	Benjamin Theurer	Proven Execution	Arca Continental's Digital Nest and AI-enabled TUALI platform are scaling generative AI across the commercial ecosystem. TUALI processes 67% of traditional-trade volume across Latin America and analyzes more than 600 store-level variables, reducing out-of-stocks by 60% and generating P\$17m of route-optimization savings.
	Ingredion	Americas Agribusiness & Mexico Consumer	Benjamin Theurer	Early Stage	Ingredion is developing AI tools that draw on customer taste profiles and internal libraries of ingredients and formulas. These tools are intended to reduce development time, improve customer adoption, and accelerate the shift toward higher-value customized solutions.
	L'Oréal	European Consumer Staples	Warren Ackerman	Proven Execution	AI is at the core of L'Oréal's operations, supported by c.€1.5bn of technology investment and a "One L'Oréal" digital backbone. L'Oréal is positioning for agentic commerce, where AI systems interpret consumer needs and identify relevant products. BETIQ delivers c.10-15% higher ROI on marketing spend, while AI-enabled formulation testing and molecular simulations accelerate discovery and compress innovation cycles.
	Unilever	European Consumer Staples	Warren Ackerman	Scaling	Technology is moving to the core of value creation at Unilever, with AI supporting agentic commerce, marketing productivity, supply-chain efficiency, and asset utilisation. Unilever is leveraging AI to improve demand forecasting, inventory management, and capital deployment, while positioning its brands to become preferred choices in AI-powered shopping and replenishment ecosystems.
	Reckitt	European Consumer Staples	Warren Ackerman	Proven Execution	Reckitt has placed digital and generative AI at the heart of its transformation, making it one of the four structural pillars of its multi-year "Fuel for Growth" programme. The most tangible progress to date has been in marketing, where the company built an internal proprietary generative AI tool that taps into tens of thousands of consumer data points and combines them with external unstructured data such as ratings and reviews, delivering a 70% reduction in concept development time in 2025, up from 60% in 2024, and twice the quality of output.
	Heineken	European Consumer Staples	Laurence Whyatt	Proven Execution	Heineken has embedded AI across its operations, using its Digital Backbone to scale GenAI and Agentic AI in marketing, sales, supply chain, and enterprise automation. Its in-house agency Freddy.ai has delivered a +36% ROI uplift and +20% improvement in advertising effectiveness, while AI tools for Revenue Margin Growth have cut insight generation time by 50% and boosted gross profit by +3-5% in key markets. The company targets €400-500m in annual gross savings from AI-enabled productivity and is restructuring its operating model to support this shift, including linking AI performance to executive incentives.
	IHG	European Leisure	Jaina Mistry	Proven Execution	IHG has partnered with Google Cloud to launch a generative AI travel planner within the IHG One Rewards app and is progressing a full cloud migration to enable AI functionality. The company lowered costs by 3% in 2025, partly driven by technology and AI.
	Accor	European Leisure	Jaina Mistry	Scaling	Accor has launched a branded ChatGPT app for searching rates and availability, is testing AI travel agents with Gemini and Mistral AI, and is upgrading its API infrastructure to support AI-driven distribution and personalisation across brands.
	Compass	European Leisure	Pravin Gondhale	Scaling	Compass has been at the forefront of technology investment in the industry. AI-driven efficiencies in procurement, menu design, and workforce management should support margins, competitive bids, and growth.
	Inditex	European General Retail	Matthew Clements	Scaling	Inditex has a data-driven operating model and a strong track record in deploying new technologies. Its integrated omnichannel model, RFID adoption, demand-forecasting systems, and real-time inventory visibility provide a foundation for AI adoption across operations and customer workflows.

Sector	Company	Industry	Barclays Analyst	Stage of AI Adoption	AI Implementation: Progress & Targets
	Next	European General Retail	Matthew Clements	Proven Execution	Next has invested roughly GBP840m in technology since FY20/21 and completed more than 70% of a major software-modernisation cycle. AI tools are already being used to improve software development productivity, supported by a modern technology stack and large pools of data and inventory.
	Dunelm	European General Retail	Richard Taylor	Proven Execution	AI contributed to GBP22m of productivity gains in FY25 through performance-marketing efficiencies and improved on-site conversion. Dunelm has deployed AI-based personalisation, with 60% of website customers seeing bespoke products, and is using AI to improve product search.
	Victorian Plumbing	European General Retail	Richard Taylor	Proven Execution	Victorian Plumbing has highlighted extensive use of machine learning AI with a current focus on productivity. AI-enhanced search functionality contributed to an increase in user conversion from 3.8% in 2024 to 3.9% in 2025.
	JD Sports	European General Retail	Richard Taylor	Scaling	JD Sports is accelerating digital transformation through the rollout of agentic AI-driven commerce capabilities.
	Frasers Group	European General Retail	Richard Taylor	Early Stage	Frasers Group announced an exclusive European launch partnership with commercetools, an AI-first commerce platform. The deployment is intended to allow shoppers to discover and purchase products from the group's retail ecosystem directly within AI shopping channels.
	Tesco	European Food Retail	Matthew Clements	Proven Execution	Tesco's combination of advanced data infrastructure, proprietary customer insight from Clubcard, and mature in-house analytics capability via dunnhumby, its wholly owned data-science business, provides a strong foundation to capture further AI-driven efficiency and personalisation benefits. The company's in-house routing tools, Tesco Hive and Bumblebee, have removed c.100k miles per week from its fleet.
	Ahold Delhaize	European Food Retail	Matthew Clements	Scaling	Ahold Delhaize has made AI and technology a core pillar of its Growing Together strategy. Its large data ecosystem and multi-banner model allow innovations to be tested within one brand and rapidly rolled out across the group, including forecasting, availability, delivery planning, and consumer personalisation.
Health care	UnitedHealth Group Inc	U.S. Health Care Facilities & Managed Care	Andrew Mok	Scaling	UnitedHealth plans to invest at least \$1.5bn in AI-related technologies in both 2026 and 2027 and expects more than \$1bn of cost savings in 2026. Optum Real, Optum Integrity One, and Crimson AI target claims, coding, and clinical analytics, supporting a multi-year opportunity approaching \$5bn of cumulative savings.
	HCA Healthcare Inc	U.S. Health Care Facilities & Managed Care	Andrew Mok	Scaling	HCA is prioritizing revenue-cycle automation and supply-chain optimization while building scheduling, staffing, and clinical-decision-support capabilities. Ambient AI is already being deployed for physician documentation to reduce administrative burden, improve clinician productivity, and support broader workflow automation.
	Eli Lilly	U.S. Is	Emily Field	Scaling	TuneLab provides biotechnology companies with access to models trained on years of Lilly research data, with the first release representing more than \$1bn in model development costs. The data flywheel should also improve Lilly's own AI and machine learning models, while the co-innovation lab with Nvidia adds further implementation capability.
	Merck	U.S. Is	Emily Field	Proven Execution	Merck's LLM-based tools have reduced the time required to create first drafts of clinical study reports from two to three weeks to three to four days. Collaborations with Mayo Clinic and Tempus combine Merck's AI and machine learning tools with multimodal clinical and genomic datasets for biomarker discovery and oncology research.
	Moderna	U.S. Small & Mid Cap Biotechnology	Eliana Merle	Proven Execution	Moderna uses AI to improve speed and efficiency developing its mRNA therapeutics. A notable case is MRNA's individualized neoantigen therapy (INT) where MRNA's proprietary algorithms allow the personalization of a cancer vaccine for each patient's individual tumor. AI allows MRNA to iteratively improve and enhance the algorithm to select the antigens to include in the therapy for each patient.
	Roche	European Pharmaceuticals, Biotechnology & Life Sciences	James Gordon	Scaling	Roche has deployed more than 2,000 NVIDIA Blackwell GPUs across the U.S. and Europe and is framing AI as a scale and infrastructure story rather than an early pilot. It is applying this capacity to modelling, data analysis, and clinical-trial processes and has integrated more than 20 AI algorithms into its digital-pathology environment.

Source: Barclays Research. For additional company detail, see Positioning for AI Disruption ([U.S.](#) and [EU](#) Editions)

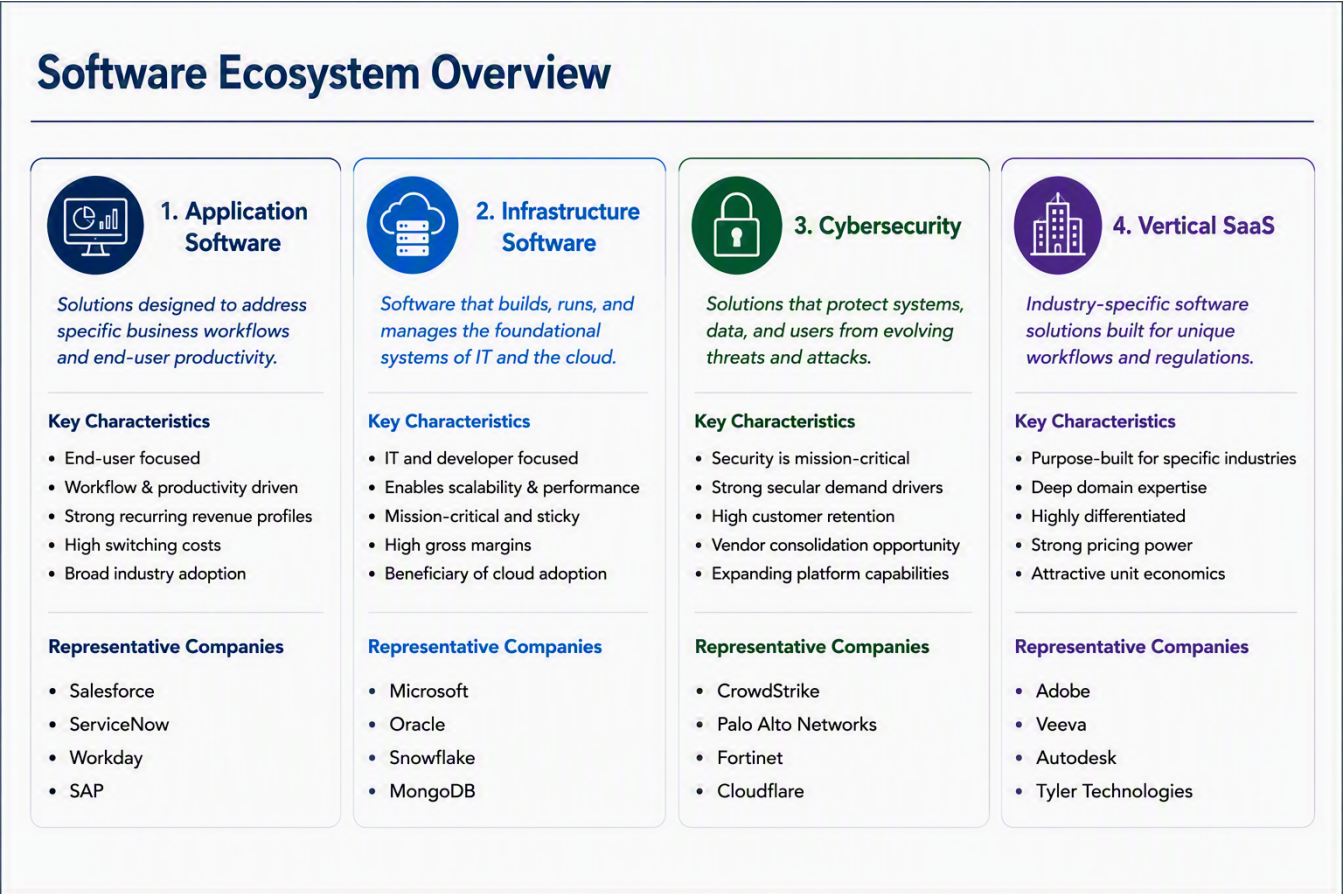
Software Vendors Helping Bridge AI Intelligence to Enterprise Outcomes

At its core, AI is software and increasing enterprise adoption of AI creates significant revenue opportunities for enterprise software vendors like Microsoft, Oracle and Salesforce that are increasingly embedding AI into their software offerings. As previously discussed, a foundational concept to appreciate is that while LLMs are advancing rapidly and becoming highly intelligent, these models lack the specific business context and proprietary data that enable impactful enterprise AI use cases. The software layer is well positioned to bridge this gap between AI intelligence and enterprise outcomes due to the rich datasets, integrations, and often mission-critical workflows that software vendors have within their platforms. Further, software tools serve as the front-end for many enterprise employees, making software platforms a natural place to deliver AI intelligence to end-users.

However, AI economics will not be shared equally in the industry, with some vendors better positioned to monetize AI while others that are unable to pivot will likely see headwinds from 1) AI's potential to address certain use cases that were previously addressed by legacy software, 2) AI's ability to lower the barriers to entry for new software creation through popular AI coding tools that are continuing to improve in capabilities and 3) AI's potential impact on seat-count growth (employment levels) that could pressure seat-based revenue models.

From a higher level, we categorize software vendors in four buckets: (horizontal) application software, infrastructure software, cybersecurity and vertical SaaS ([Figure 18](#)). Broadly, we see infrastructure and cybersecurity vendors as more clearly positioned to benefit from accelerating AI adoption, with application software and vertical software to benefit in aggregate but with greater dispersion amongst winners and losers.

FIGURE 18. Software Ecosystem Overview

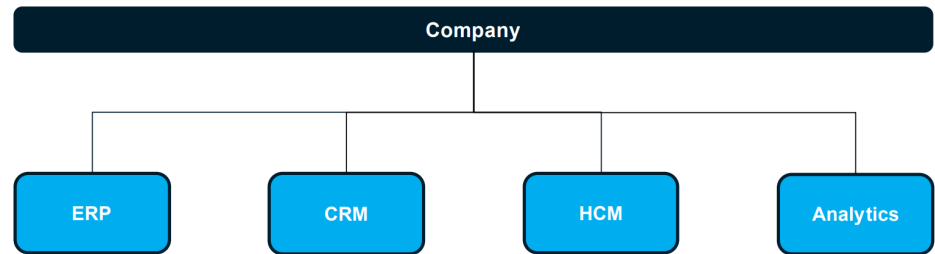


Source: Barclays Research

1) Horizontal Application Software

The application software space is likely most familiar to investors and includes software solutions designed to address specific business workflows (e.g., workflows in the ERP, CRM or HCM spaces) and improve end-user productivity (e.g., Microsoft's Office suite). Horizontal application software vendors like Salesforce, Workday and ServiceNow deliver software solutions that are generally customer industry agnostic, unlike vertical SaaS vendors that primarily cater their solutions to a particular industry or subsegment of the market. Since the early 2000s, the application space has evolved from on-premises deployments and perpetual licenses to primarily a cloud delivery model (SaaS), where the applications themselves are hosted in public or private clouds and run/maintained by the SaaS provider. Now, we are seeing another evolution in the SaaS market with the proliferation of LLMs and AI agents, which application vendors are increasingly leveraging within their platforms to deliver better services to customers.

FIGURE 19. Main Horizontal Application Software Suites



Source: Barclays Research

Key Risks Driving the Debate

Despite the opportunities for application software vendors to capture value from AI adoption, we expect AI economics to be shared unevenly by market participants, and note that AI has created considerable debate for the app software space. Below, we outline the key factors that are creating this debate.

- **Higher unemployment from AI impacting seat-based models:** The first area of concern regarding AI's potential negative impact on enterprise application software vendors revolves around the fear that AI could significantly displace employees and lead to workforce reductions, negatively impacting enterprise software demand, particularly for vendors with seat-based revenue models.
- **Competitive threats from AI-native startups:** The next point of debate is tied to new AI-native market entrants that leverage AI to help erode incumbent vendors' market share. There are two nuances here: 1) AI can help new entrants more rapidly create (vibe code) competing solutions for existing use cases, and 2) incumbents could leverage AI to create new/better solutions for existing use cases.
- **Monetization challenges:** The third key discussion area revolves around the potential issue of monetization challenges, particularly if aspects of GenAI solutions get commoditized. Vendors may fail to capture meaningful GenAI revenue and inference costs incurred just to compete could pressure gross margins.

Aspects Positioning Application Software Vendors for AI Monetization

Despite these risks, we see software vendors well positioned to benefit from AI advancements and the adoption trends outlined earlier in this report. Below we outline the key underlying reasons why SaaS vendors will be a vector with which AI value accrues over time:

- **Data gravity and metadata advantages:** Deep integration of enterprise data and metadata creates defensible moats and hence, enterprise software app platforms often have important data and context that is needed for effective AI applications. See more on this topic: [rAlmo's AI Series - Vol. 2: The Data Gold Rush](#); 9/11/2023.
- **Embedded workflows and automations:** Mission-critical processes are deeply tied to existing platforms and hence, become a natural place to run AI-augmented workflows.
- **Integration ecosystems:** Strong connectivity across enterprise systems via out of the box connectors and marketplaces limit switching.
- **Deployment flexibility:** Some customers have unique deployment needs. Options for on-prem, hybrid, and cloud deployments meet diverse customer and workload needs, while AI natives may be cloud-only.
- **Security and compliance standards:** Enterprise-grade security remains a differentiator for incumbents, with certain certifications (FedRamp) taking time. Many customers require multi-region deployments, which often face different regulatory requirements.
- **Distribution channels / professional services:** Most enterprise SaaS companies have an ecosystem of partners that support the implementation, education, and ongoing utility of tools that make adoption more seamless. Healthy partner ecosystems help drive top-of-funnel demand and vendor mindshare.

Well Positioned Vendors

Below, we outline a set of vendors in our enterprise SaaS coverage universe that are well positioned to benefit from AI adoption and have already demonstrated success in monetizing AI.

FIGURE 20. Application Software Vendors Demonstrating AI Monetization

Company	Notable AI Products / Offerings	AI ARR / Revenue Metric	AI Bookings / ACV / NNACV Impact	AI Customers	AI Usage Metrics
Atlassian	Rovo, Teamwork Graph, Duo Agent Platform	Rovo adopters grow ARR at >2x the rate of non-adopters		>80% of Fortune 500 companies now use Rovo (4Q26)	Rovo-assisted actions grew 50% QoQ (4Q26), Agentic automations in Service Collection have increased 3x over the past 6 months (May 26), Teamwork Collection customers deploy 2x as many active agents as standalone customers and use 2x as many AI credits per user
Braze	BrazeAI, Agent Console, Decisioning Studio			AI tool adoption increased 900 bps QoQ among large customers, Paid adoption of AI tools reached ~1/3 of large customer cohort in 2Q27	
Five9	Intelligent CX Platform, Five9 Genius AI, Voice AI Agents	\$150M+ AI ARR run-rate (2Q26), AI revenue grew 78% Y/Y in 2Q26, accelerating from 68% in 1Q	AI revenue makes up ~20% of enterprise NNACV bookings, 100% YoY growth in Enterprise AI Bookings 4Q25		
Gitlab	Duo Agent Platform (DAP), GitLab Orbit, GitLab Flex	Duo Agent Platform (DAP) Paid CRR grew ~50% QoQ (2Q27)		DAP attached to 4 of top 10 deals (1Q27)	Since opening the beta in June, more than 2,200 organizations have enabled Orbit indexing, up 70% in four weeks. Customers have generated more than 170,000 queries
Intuit	Intuit Intelligence, AI Agents, Intuit Accountant Suite			>75% of Enterprise Suite customers engage with Intuit Intelligence monthly (Sept 2026)	400M transactions categorized by AI monthly (90%+ of all transactions) (Sept 2026)
Microsoft (Copilot)	M365 Copilot, Copilot Studio			M365 Copilot reached >30M paid seats (FY26), Copilot paid seats grew >160% Y/Y (FY26)	
monday.com	Monday Agents, Sidekick, Monday Vibe	Monday Vibe is the fastest product to surpass \$1M ARR; ARR from AI products doubled from 1Q, representing 17% of net new ARR		Over 250,000 customers worldwide use monday.com to bring people, workflows, and AI agents together on one flexible platform	1.70M+ monday agent interactions; 98M+ AI block actions (2Q26)
Oracle (Apps)	AI Agents (Fusion & Vertical)			2,400 customers are live on embedded AI within industry applications (Oct 25)	900bn Fusion Tokens Consumed in the quarter, 2,300 Agents in Production (up 90% QoQ), 3.5mn Agent executions in the quarter (1Q27)
Salesforce	Agentforce, Data 360, Claudeforce	\$3.90B combined ARR for Agentforce + Data 360 (2Q27); Agentforce ARR grew >240% Y/Y; Data 360 ARR grew >200% Y/Y	Bookings for Agentforce 1 Edition and for apps "more than doubled quarter over quarter" (2Q27)	29,000 Agentforce deals closed to date (FY26)	3.20B Agentic Work Units (AWUs) in 2Q27

Company	Notable AI Products / Offerings	AI ARR / Revenue Metric	AI Bookings / ACV / NNACV Impact	AI Customers	AI Usage Metrics
SAP	Joule, Joule Studio, Autonomous Suite	€2B efficiency run rate expected by end of 2028	AI and Data Cloud embedded as key pillars in >90% of top 50 deals (2Q26)	>2,000 AI Factory customers (2Q26)	>110k Joule Work Desktop users
ServiceNow	Now Assist, AI Control Tower		ServiceNow AI ACV exceeded \$1B in 2Q26; on track for \$1.50B target		AI Agent assist consumption increased >55x since May 2025 (as of 3Q25), Agentic deployments of ServiceNow AI increased 9-fold in 9 months
Workday	Agentic AI, Developer Agent, Organic Agents	~\$600M ARR from Agentic AI (2Q27); Agentic AI ARR grew >200% YoY; Organic agent customers up >35% QoQ	AI products drove >\$100M new ACV, representing >25% of total new ACV in 2Q27	>5,500 customers using at least one organic agent (as of 8.15.2026)	

Source: Company data and disclosures, Barclays Research

Additional Resources

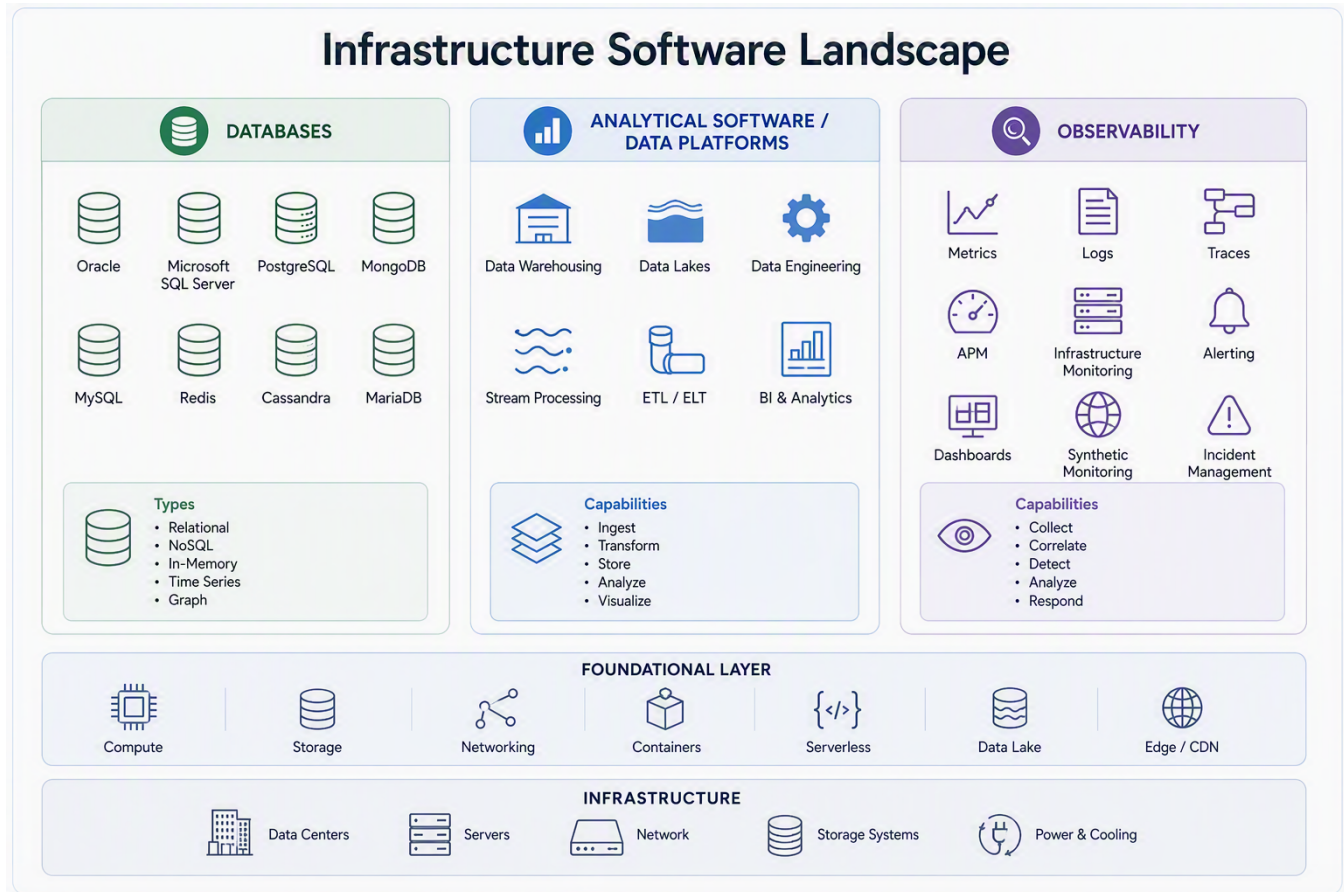
To learn more about the application software layer within the context of AI adoption, see the below reports.

- [U.S. Software: Software Is Not Dead, Just Changing](#) (2/10/26)
- [U.S. Software: rAlmo's AI Series - Vol. 3: Agents Take the Driver's Seat Alongside Copilots](#) (11/11/24)
- [U.S. Software: rAlmo's AI Newsletter #113: Headless SaaS Offerings - The Beginning Of The End?](#) (6/8/26)
- [U.S. Software: The Application Software Handbook](#) (12/6/21)

2) Infrastructure Software

Within enterprise software, we see infrastructure software vendors as well positioned to benefit from AI adoption trends. Unlike application software providers, infrastructure software vendors deliver solutions that help build, run and maintain the foundational systems that underpin a companies' IT deployments. For example, database vendors like Oracle and MongoDB, observability software vendors like Datadog and Dynatrace, and analytical software vendors like Snowflake fall under this category. Infra software companies often also have different revenue models vs. app vendors, where revenue is more commonly tied to some proxy of consumption, usage or capacity with the underlying drivers often being usage of compute and/or storage. Vendors in this space also offer application development platforms that are increasingly AI focused, allowing organizations to more easily build AI applications. Broadly, this revenue model positions the infrastructure software layer well as AI and agentic AI adoption is driving significantly more activity within the infrastructure layer.

FIGURE 21. Higher Level Infrastructure Software Overview



Excludes cloud service providers and cybersecurity vendors that are discussed in following sections.

Source: Barclays Research

Key Risks Driving the Debate

We note relative to application software, infrastructure software vendors have seen less scrutiny around the AI benefits/disruption debate. Below, we outline some of the key risks related to infrastructure software vendors in the context of AI and the AI adoption trends outlined earlier in the report.

- Competition concerns:** An important aspect to note is that many infrastructure software areas have open-source equivalents that create competition for the more out-of-the-box solutions that software vendors provide. For example, customers have choice between using proprietary databases from vendors like Oracle, MongoDB and the hyperscalers, or using open-source databases such as Postgres, MySQL or Apache Cassandra. As AI adoption continues, vendors must win developer mindshare in order to monetize AI's consumption of infrastructure software services.
- Technology/Abstraction risk:** Infrastructure software vendors typically see greater technology risk relative to application software companies as advancements in computing architectures, databases, networking, storage, developer frameworks, and AI systems can change the requirements of the infrastructure layer directly, while the app layer typically sits closer to the user and workflow, where outcomes and business processes drive more of the decision framework vs. the underlying architecture. AI is increasingly abstracting certain areas of infrastructure software (e.g., data engineering workflows) and AI is increasingly making decisions of what infrastructure to use (e.g., an AI agent provisioning a vector database).

- **AI's ability to lower switching costs and commoditize certain areas:** The market is also seeing rise of more powerful AI migration tools that help organizations move from one software provider to another provider. Advancements in these tools could reduce vendor lock-in from high switching costs and have implications for gross retention for some infra software vendors.

Aspects Positioning Infrastructure Software Vendors for AI Monetization

Below we outline the key underlying reasons why infrastructure vendors are well positioned to capture value from AI adoption over time and the trends outlined within the note:

- **Data gravity and metadata advantages:** Infrastructure software vendors also benefit tremendously from data gravity and metadata advantages, but often from a slightly different angle than application software peers. App software vendors data moats are relatively centric to the workflows and use cases being served, while infra software vendors importantly provide various data management and observability functions to customers. Hence, many of these vendors sit on top of rich sets of data within the database and data platform services that these vendors provide, hence, providing a natural place to provide context to AI systems (see [U.S. Software: Raimo's Roundup: Vector Databases – A New Tool for the Gen AI World](#), 8/4/23).
- **Well positioned to benefit from explosion of new software creation:** A trend we are seeing in the market from the advancements and popularization of AI coding tools is an explosion in the amount of software and applications that are being created (see [U.S. Software: rAlmo's AI Newsletter #115: Unpacking the AI-Driven Software Explosion](#), 6/22/26). Many areas within infrastructure software benefit from more software/application development. For example, new code needs to be stored, managed, tracked and collaborated on (code repositories, MSFT/GTLB), new applications need databases (MDB, ORCL), which need to be observed and monitored (DDOG, DT), driving more demand and usage for these types of offerings.
- **Integration and data processing ecosystems:** Strong connectivity across enterprise systems via out of the box connectors and marketplaces limits the value proposition of switching platforms/providers. Additionally, some vendors in the data analytics space provide data processing and data analytics solutions, which can serve as an input for agentic/AI workflows. We see this layer (e.g., data platforms like Snowflake, iPaaS solutions like Workato (private, not covered) increasing in importance as agents need high quality data and integrations to perform tasks accurately.
- **Critical Role in AI Security and Compliance:** Enterprise-grade security remains a differentiator for incumbents, with certain certifications (FedRamp) taking time. Infrastructure software companies also play an important role in data governance, which is critical for secure enterprise AI.

Well Positioned Vendors

Below, we outline a set of vendors in our infrastructure software coverage universe (excluding the compute layer, which is discussed explicitly in a following section) that are well positioned to benefit from AI adoption and have already demonstrated success in monetizing AI.

FIGURE 22. Infrastructure Software Vendors Demonstrating AI Monetization

Company	Notable AI Products / Offerings	AI ARR / Revenue Metric	AI Bookings / ACV / NNACV Impact	AI Customers	AI Usage Metrics
CoreWeave	CoreWeave AI Cloud	Surpassed \$10bn in annualized revenue run-rate in 2Q26, growing ~112% y/y, virtually all related to AI infrastructure revenue	Surpassed \$104bn in total RPO in 2Q26 for growth of ~246% y/y, virtually all related to AI infrastructure bookings		
Datadog	Bits AI, LLM Observability, Agent Console	AI-native cohort contributed "high single digit" percentage to Y/Y revenue growth, AI customers represent ~80% of total ARR (1Q26)		>6,500 customers using one or more AI integrations (1Q26), Serving about 650 AI native customers (4Q25)	
DigitalOcean	AI-Native Cloud, Inference Engine	\$234M AI Customer ARR and grew 212% YoY (2Q26), 85% of AI ARR comes from inference and core cloud vs. bare metal	Total RPO reached \$894 million in 2Q26, a 12x increase from \$75 million in the prior year period, heavily influenced by AI		Early Inference Engine customers drove their total token consumption up approximately 30x in the last 60-days (2Q26)
Elastic	Elasticsearch, Search AI Platform, AI Assistants			>2,450 customers on Elastic Cloud using GenAI (2Q26)	
IBM	Watsonx ecosystem, IBM Granite, IBM Bob, Project Lightwell	IBM signed a \$240 million agreement with Together AI (private, not covered) to deploy a large-scale inference cluster using NVIDIA HGX B300 systems on IBM Cloud by early 2027			
Microsoft (Azure)	Azure Foundry, Azure AI	Azure revenue surpassed \$100bn for the first time in FY26, Azure growth of 43% in 4Q26	Commercial RPO grew from \$392 billion in 1Q26 to \$678 billion by 4Q26, heavily influenced by AI	100,000 Azure AI Foundry Customers as of 3Q26	
MongoDB	Atlas Vector Search, MAAP, Voyage AI Embeddings	~30% of Atlas ARR comes from customers with at least one AI use case (Sept 2025)		Voyage customers roughly doubled QoQ for two consecutive quarters (2Q27)	48% of \$100k+ Atlas customers use 2+ features, driven by search adoption (Sept 2025)
Oracle (Infra)	Oracle AI Cloud	\$6.50B CPU & GPU Revenue which grew 151% Y/Y (1Q27)	Booked >\$30bn of additional cloud contracts in 1Q27, \$664bn RPO Total, up from \$138bn exiting FY25, heavily influenced by AI	300,000 GPUs delivered to AI Cloud Customers (1Q27)	AI Infrastructure Utilization reached 97.90% (1Q27), 850 MW of capacity added in 1Q27 alone
Snowflake	Cortex AI, CoCo (Cortex Code), CoWork	\$100mn in AI consumption revenue run-rate as of 3Q26	Noted ~50% of bookings were influenced by AI in 3Q26	CoCo surpassed 9,100 accounts (2Q27), adding 2,000 in the quarter, while CoWork expanded to 5,800 accounts	
UiPath	UiPath Maestro, IDP, Communications Mining	Noted >\$250mn in AI ARR during its Sept 2026 Investor Day, growing over 50% y/y		Noted customers that have adopted both the platform and AI in the \$100k-\$1mn cohort saw >120% NRR, vs. 109% for total company (2026 Investor Day)	

Company	Notable AI Products / Offerings	AI ARR / Revenue Metric	AI Bookings / ACV / NNACV Impact	AI Customers	AI Usage Metrics
WhiteFiber	WhiteFiber AI Cloud Services, WhiteFiber Colocation Services	Noted >\$250mn in AI ARR during its Sept 2026 Investor Day, growing over 50% y/y		Noted more than \$550mn in new AI cloud services deals between May and August 2026, Signed \$865mn TCV in December 2025	

Source: Company data, Barclays Research

Additional Resources

To learn more about the application software layer within the context of AI adoption, see the below reports.

- [U.S. Software: rAlmo's AI Newsletter #115: Unpacking the AI-Driven Software Explosion \(6/22/26\)](#)
- [U.S. Software: rAlmo's AI Series - Vol. 4: The Emerging Big Data Platform Battle \(12/9/24\)](#)
- [U.S. Software: rAlmo's AI Series - Vol 5: The Platform War for Agents \(9/19/25\)](#)
- [U.S Software: Observability Industry Review: Volume 16 - SRE AI Agents \(2/4/26\)](#)
- [U.S. Software: rAlmo's AI Newsletter #117: Checking in on AI's Impact on Developer Productivity \(7/6/26\)](#)

3) Cybersecurity

[Saket Kalia, CFA](#) BCI, US

Cybersecurity software encompasses the platforms and tools organizations use to protect networks, endpoints, cloud infrastructure, applications, identities, and data from cyber threats. Traditionally, cybersecurity vendors focused on preventing unauthorized access, detecting malicious activity, and helping enterprises respond to security incidents. As businesses increasingly digitize operations and move workloads to the cloud, cybersecurity has evolved into a mission-critical layer of enterprise software, responsible for safeguarding an organization's most valuable digital assets. AI adoption is further expanding this mandate by introducing new assets that require protection, including AI models, agents, prompts, training data, vector databases, and AI-generated code.

Cybersecurity software vendors are particularly well positioned to benefit from AI adoption because AI simultaneously increases both the number of systems that must be secured and the sophistication of potential threats. Enterprises are now seeking solutions to govern AI usage, secure AI agents, monitor emerging risks such as prompt injection and data leakage, and defend against increasingly automated attacks powered by AI. At the same time, security vendors can leverage AI within their own products to improve threat detection, automate incident response, and reduce the burden on overstretched security teams. This creates a powerful demand cycle: AI expands the attack surface while also making advanced security capabilities more valuable, driving incremental spending on both traditional cybersecurity platforms and emerging AI security solutions.

Key Risks Driving the Debate

- **Competitive risk from AI vendors:** AI vendors have released tools in certain parts of the security market with Anthropic's Claude Code Security and OpenAI's Codex Security, which begs the question of whether any security tools or security markets could be displaced by AI vendors. That said, we note that horizontal platforms have historically had mixed success in security at best because the margin for error is so low, pushing most buyers towards security specialists, which is why we believe the leading AI vendors will focus on partnering with security vendors. Zooming in, we think security markets with valuable real estate like network security and endpoint are likely more insulated from this disruption risk, while areas with less of a technical moat, like code security, may be more at risk.
- **Prioritization of enterprise security investments in the face of AI risk:** AI is changing the security landscape in several ways, which begs the question of how enterprises will prioritize their security investments and which areas within security will benefit first. The Mythos-

driven threat is top of mind for enterprises as it enables threat actors to create more sophisticated attacks and accelerate the rate of attacks, which could drive demand for vulnerability management and patching tools, network segmentation tools, and cyber resilience tools. Separately, broader adoption of AI tools and agents drives a need for data security and identity security tools as enterprises ensure that sensitive data isn't leaked, control what data AI tools and agents have access to, and manage the identities and entitlements of agents. Greater AI and agentic traffic could also drive greater investments in endpoint security, network security, and CDN, so the question of which area of security could see the biggest benefit from incremental spending dollars near-term versus long-term and which vendors are best positioned is well debated.

- **Platform versus point product debate in the AI era:** The debate around vendor consolidation and whether enterprises prefer to have the "best-of-breed" point solution or take the platform approach by standardizing on fewer vendors is not new to the security space - that said, we believe this debate becomes more important as growing AI adoption expands the attack surface and the release of more advanced AI models accelerates the pace and sophistication of cyber-attacks. In this AI-era, we believe major security platforms like Palo Alto Networks and CrowdStrike are best positioned as it becomes increasingly important for security tools across the stack to work together, giving an enterprise unified visibility so it can identify and remediate AI-related threats at pace.

Aspects Positioning Cybersecurity Vendors for AI Monetization

- **The release of advanced AI models is enabling faster and more sophisticated attacks as threat actors and autonomous AI agents can find and exploit vulnerabilities at machine-speed:** The release of more advanced, Mythos-level AI models is driving a greater sense of urgency for enterprises to invest in their security architectures as these models can find and exploit vulnerabilities at machine-speed. This puts more powerful tools in the hands of threat actors and also creates a risk as AI agents are acting autonomously to gain unauthorized access to enterprise systems and find/exploit vulnerabilities – we've seen several examples of this with the [OpenAI and Hugging Face incident](#), [Anthropic Claude incidents](#), and [Google Gemini incidents](#). This is driving incremental spending on security tools, and we believe areas like vulnerability and patch management, network segmentation, and cyber resilience are particularly well positioned to benefit.
- **AI adoption drives a need for data security and identity security:** As enterprises broadly deploy AI applications and AI agents across their organizations, we believe three goals are top of mind: (1) ensuring sensitive data is not accidentally leaked through prompts in AI models; (2) controlling what data AI tools and agents have access to; (3) managing the identities and entitlements of agents. As a result, we believe the growing use of AI tools and agents in the enterprise creates a near-term revenue opportunity for data security and identity security vendors with tools that address these needs.
- **Growing AI traffic and broader use of AI tools and agents could drive more investments in network and endpoint security:** For endpoint security, as more AI agents, browsers and applications run on the endpoint, this positions the endpoint as a more strategic control point rather than just an enforcement layer. From a network security perspective, we wonder if growing AI and agentic traffic could lead to greater investments in network security architectures over time, and whether the build out of AI infrastructure could drive demand for firewalls in the near-term.
- **AI security is a direct revenue opportunity for security vendors as enterprises secure the AI models, applications, and agents deployed across their organizations:** Several security vendors in our coverage have introduced AI security tools both to secure AI usage and to

improve productivity within the SOC – examples include Palo Alto's Prisma AIRS, CrowdStrike's Falcon AIDR and CharlotteAI, Okta for AI Agents, Cloudflare's AI Security Suite, and Zscaler's Security for AI portfolio. While it's still early for many of these solutions, some have been showing notable traction and we would expect contributions to materialize as deployments scale. We believe platform vendors with AI security offerings like PANW and CRWD are well positioned as enterprises will likely turn to trusted platform vendors to secure their AI estate, in our view.

Well Positioned Vendors

- Palo Alto Networks (PANW) – a platform vendor capitalizing on AI-driven network evolution.** The use of new AI tools, applications, agents, and workloads could drive an evolution in an enterprise's network, which along with growing AI traffic and the build out of AI infrastructure could drive demand for network security tools like PANW. At the same time, advanced AI models are putting powerful tools in the hands of attackers which could drive security spend for network segmentation tools and for platform security vendors in particular. In addition, enterprise adoption of AI expands the attack surface creating the need for new security tools, and we believe platform vendors like PANW are well positioned to capture this demand – PANW's AI security offering Prisma AIRS drove ~\$120M in ARR and had over 800 customers as of the end of FY26 (fiscal year ended July 2026).
- CrowdStrike (CRWD) – another scaled security platform with roots in endpoint.** Similar to PANW, CrowdStrike has established itself as a large-scale, unified security platform, and we believe the company is on track to become durable, mission-critical infrastructure for both securing and enabling global AI adoption. As enterprises increasingly deploy AI tools, browsers, and agents, much of this activity runs directly on user devices, elevating the strategic importance of where and how security controls are enforced. In this context, the endpoint serves as a critical anchor point within a broader platform architecture, positioning CrowdStrike to naturally benefit. Zooming out, similar to PANW, CRWD's broader platform is likely to benefit from the expanding attack surface and growing demand for new security tools. Notably, CRWD has been actively building out its platform to secure AI-native workflows, including the acquisition of Pangea in 2025, which added AI Detection and Response capabilities for Falcon Guardian, and CharlotteAI, CRWD's genAI assistant for security operations, which saw usage increase dramatically 6x y/y as ARR more than tripled in 4Q26.
- Cloudflare (NET) – the control plane for agentic internet.** Cloudflare breaks up its products into "acts" with Acts 1, 3, & 4 best positioned to benefit from AI-driven trends. Act 1 is Cloudflare's CDN product suite and benefits as AI agents generate significantly higher request volumes than human users, and the company monetizes traffic whether it originates from humans or agents. On Act 3, we see Cloudflare as an AI beneficiary with inference use cases becoming increasingly latency-sensitive, which is driving demand for Worker's edge compute. Also withing Act 3, the Workers developer platform is seeing rapid growth in developer count. Act 4 is Cloudflare's answer to the problems created by AI search summaries capturing traffic without sending monetizable traffic back to those publishers and allows content creators to block or charge AI scrapers for access to their content. In our view, Cloudflare's is well positioned to benefit with a flywheel of product suites feeding into one another as AI/agent use grows.
- Varonis (VRNS) – the security layer for AI tools.** The rising use of AI tools in enterprises presents a critical security question: How do we lock down an organization's sensitive data to ensure users and agents only have access to the data they need? As a result, data security is becoming an increasingly important component of an enterprise's security stack, where Varonis is a market leader with dedicated Microsoft 365 Copilot and ChatGPT Enterprise

SKUs. Varonis also recently acquired AllTrue (now Varonis Atlas), to inventory and monitor an organization's AI tools, workloads, models, agents, etc, and provide guardrails around it all. As a result, Varonis can monetize the AI opportunity both through its dedicated AI Security SKUs and also as AI expands the TAM in data security, creating a broader opportunity for VRNS to go after.

- **Okta (OKTA) – identity vendor for secure agentic deployment.** Enterprises are increasingly deploying agents, but in order for an agent to be more useful than just a general chatbot, it requires access to internal data, applications, tools, etc. The problem is that not every agent needs or should have access to all internal tools, similar to how human employees are granted different levels of access based on their role. Okta's agentic AI offering, Okta for AI Agents, is meant to address this by applying the company's expertise in human identity security to these new agentic identities to let enterprises control what agents can access, monitor activity, and tie agents back to a human identity for accountability. We think Okta is well positioned to benefit from the proliferation of agents in the enterprise, as it expands the number of identities that need to be secured and governed which creates a larger opportunity for monetization given how crucial identity is for secure agent deployment.

4) Vertical SaaS

[Saket Kalia](#), CFA BCI, US

Vertical SaaS software consists of industry-specific applications built to serve the unique workflows, regulatory requirements, and operational needs of a particular end market, such as healthcare, construction, legal services, field service, hospitality, or financial services. Unlike horizontal software that serves broad functions across industries, vertical SaaS providers embed deep domain expertise directly into their products, often becoming the system of record and operational backbone for their customers. This specialization creates strong customer relationships, valuable industry-specific datasets, and workflows that are difficult to replace. AI is increasingly becoming part of customer conversations around vertical solutions, helping expand both platform adoption and strategic value.

Vertical SaaS vendors are particularly well positioned to benefit from AI adoption because they possess the proprietary data, workflow context, and domain expertise needed to deliver highly differentiated AI-powered solutions. Rather than offering generic AI capabilities, these companies can embed AI directly into industry-specific workflows, automating tasks that previously required human labor and improving decision-making within the context of a customer's day-to-day operations. As AI evolves from a productivity assistant to an execution engine, the value increasingly accrues to platforms that own the workflow, the data, and the customer relationship. This creates opportunities for higher revenue per customer, deeper product penetration, and expansion into adjacent services, while strengthening the competitive moat of leading vertical SaaS providers.

As vertical SaaS vendors introduce new agentic AI capabilities into their customers' existing workflows, many vendors are able to leverage years' worth of customer data to create "deterministic" models that can tailor specific queries and workflows to those industries' unique use cases. These AI tools can then pass on other queries to the "non-deterministic" frontier LLM providers to handle work that is broader in scope or requires more input. The benefits of this are twofold: (1) for the vertical software vendor, this lowers the cost of providing these AI tools to customers, leading to a potential gross margin profile similar to that of the existing SaaS business; (2) for the customers, this provides more tailored responses based on vetted and compliant industry data that can garner a higher degree of trust.

Key Risks Driving the Debate

- **Competitive risk from AI vendors:** Frontier AI Labs and other AI vendors have released tools that have similar capabilities to some verticalized software vendors that we cover. In the creative market, we have seen vendors like the frontier labs and startup vendors release text-to-image and text-to-video functionality at a low cost to customers, and, arguably, this has begun disrupting the traditional creative software industry. Second, with model releases from Open AI and Mistral, we have fielded more questions on competitive moat in the industrial software market. Our belief is that while there is less insulation in areas like creative software, we believe companies that serve highly regulated fields like Architecture, Construction, Manufacturing, and Civil Engineering are more insulated due to the complexity and need for deterministic outcomes of these industries.
- **Technological maturity of endmarkets:** While many vertical companies offer AI solutions or SKUs, it is important to consider the technological maturity of the endmarkets these vendors serve when considering adoption. For example, the construction industry is one of the least digitized industries in the world, only in front of agriculture and fishing, and much of the market has yet to adopt vertical software altogether. Additionally, state and local governments, while digitized, are slower to adopt new technologies. An example being that while much of the software that we use today is SaaS, this industry still operates analog and on-premise software, despite availability of cloud tools. Other heavily regulated industries like financial services and healthcare have a similar tendency to prefer on-premise hosting of their data along with stringent regulatory requirements to meet with any software vendors they do business with, so we wonder if broader technological maturity could be a bottleneck to AI adoption.
- **Horizontal vendors with larger war chests vs. specialized vertical vendors:** Vertical SaaS has been a heavily debated sub-sector in 2026 as investors pondered whether these software companies are defensible, or whether the workflows they offer customers can be easily replicated either by horizontal providers that are leaning into AI more aggressively or by customers who could “vibe-code” their own versions. Over time however many of these vendors have not only continued to add new customers by demonstrating their existing value propositions, but have provided an on-ramp to new AI tools of their own that lean on existing vendor relationships and keen focus on vendors with regulatory compliance built into their solutions. Vertical software vendors for certain industries are subject to regulations including inspections of their code by regulators themselves, and other scenarios can also prove to be key value adds such as professional services teams in an outage or downtime scenario.

Aspects Positioning Vertical SaaS Vendors for AI Monetization

- **AI helping to address the labor shortage in many industries:** Verticals such as construction and government notably are endmarkets that have a shortage of labor, which Procore recently sized the US construction labor shortage to be ~350k. Because of the labor shortage, we believe customers of these endmarkets could increasingly leverage AI to complete time-consuming manual tasks. For example, Procore customers have utilized AI that powers robotics to perform daily checks on material supply, Autodesk customers have leveraged AI to document changes to architectural models, and Tyler Technologies customers have leveraged AI to automate application review for licensing and permitting and data entry into courts systems.
- **Deep domain expertise and unavailability of viable training data:** Frontier Labs that created tools like ChatGPT or Claude used data across the internet to train their models into efficient AI tools. While this may address more general repeatable tasks quite well, specialized and complex tasks may not have data across the internet for LLM’s to train on. This is an area where we think vertical vendors of specialized fields like Manufacturing,

Construction, Architecture, and Civil Engineering have a right to win. Detailed architectural drawings, construction plans, or manufacturing designs are not widely available for the public to use, and thus these vendors that serve these endmarkets have a data advantage, in our view. We ultimately believe that customers of these industries will partner with those companies that have deep domain expertise as they look to adopt AI.

Well Positioned Vendors

- Autodesk (ADSK) – a design, make, and operate platform vendor with valuable relationships and data.** Companies across industrial endmarkets have long felt labor as the key bottleneck of productivity, but with the release of specialized AI solutions for design, make, and operate specific tasks, we think Autodesk will be able to help its customer base and endmarket through the ongoing labor shortages. We believe Autodesk is one of the best positioned vertical software vendors for AI adoption because the data that it possesses is highly specialized, rare, and valuable as well as its deep domain expertise across its endmarkets. We think three solutions the company just announced within Autodesk Assistant, Autodesk Context for retracing decisions and project implications, Assistant Builder for connecting in-house and third-party models, and AI Orchestrator for continuous model evaluation for specific tasks, are all relevant to this idea.
- Procore (PCOR) – another specialized vertical vendor with construction specialization.** Procore is a vertical software that exclusively serves the construction market, and while Autodesk has customers across design, make, and operate, we believe the themes we mentioned are also relevant to Procore. An example that resonates with us is of a customer who took a video of a column in a building, the tool found the column in the drawings, and then was able to identify areas of improvement/differences - it then created work orders for these improvements, and notified all the parties that were impacted by these changes. The construction labor shortage has limited the industry from growing faster, and many construction companies have years-long backlogs – we think real examples like these show ROI and these tools can become potential co-workers.
- Veeva (VEEV) – industry cloud for life sciences with a growing AI roadmap.** Veeva is a vertical software vendor that provides two major categories of software for global pharma & biotech companies: (1) Commercial Solutions including Veeva CRM (currently migrating customers to Vault CRM, a newer iteration) and other tools for optimizing marketing spend and sales rep interactions; (2) R&D Solutions to automate the documentation of clinical trials, quality assurance, regulatory submissions, and safety events. Veeva is undergoing a phased rollout of Veeva AI agents that sit within the existing Vault platform here in 2026 that will speed up human labor required to complete certain tasks, and also introduced Veeva Falcon in mid-2026 which is an agentic platform for full automation of certain tasks that were previously done by humans. This gradual approach of validating AI use cases through early adopter customers provides a viable on-ramp to new AI tools for a customer base that prioritizes regulatory compliance and accuracy.

AI Labs and Model Providers Building the Intelligence

Frontier labs are currently capturing a meaningful share of AI economics. On balance, we see solid profitability, even after accounting for training costs, though margins vary by business mix and revenue share agreements ([link](#)). For the biggest two labs, which we denote as Lab A and Lab B, we estimate that subscription and direct API generates ~70% inference margins, while indirect API is moderately lower (~40/50%) given higher revenue sharing with partner clouds. Another variable, Lab B has a meaningful strategic partner revenue share agreement, while Lab A does not.

FIGURE 23. AI Lab Economics Vary By Business Mix, Strategic Partner Revenue Share

AI Lab Economics	Lab A	Lab B
Customer Spend	\$100	\$102
Revenue	100.0	100.0
Subscription	30.0	80.0
Direct API	45.0	10.0
Indirect API	25.0	10.0
Costs	35.0	52.0
Indirect Cloud Fees	5.0	2.0
Strategic Partner Revenue Share	0.0	20.0
Inference Compute Cost	30.0	30.0
AI Lab Paid Inference Profit	65.0	48.0
<i>% Margin</i>	65%	48%
Model Production Training Cost	10.0	10.0
AI Lab Gross Profit (After Training)	55.0	38.0
<i>% Margin</i>	55%	38%

AI Lab ARR Definitions Are Different

~10% of Total Training Costs Allocated to Final Model Production

Source: Barclays Research

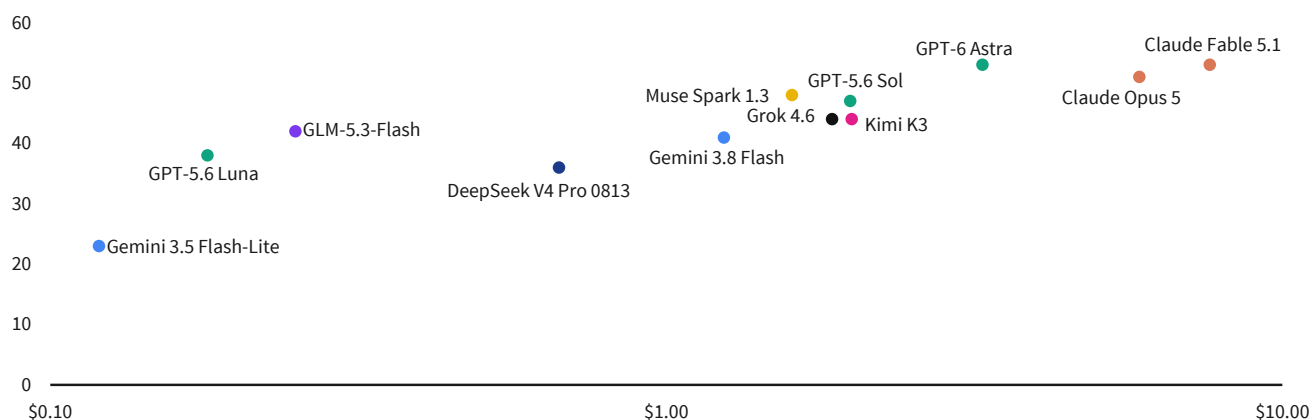
Frontier labs are signaling the need to pace development, and we'd expect renewed efforts to monitor cyber-qualified models. This is a practice called synchronous chain-of-thought monitoring, and could increase compute usage by ~20% ([link](#)). To the extent this materializes, this would be a "tax" on lab inference margins. It is possible that the labs may pass on higher costs by further increasing token prices, but competition is growing on that front.

FIGURE 24. "Monitoring" Spend Could Add \$76B To Industry-Wide In '28

Compute Overhead Breakdown (\$B)	2026E	2027E	2028E
Total Base Compute Cost	\$113B	\$246B	\$426B
Training	66	132	210
Inference	47	114	216
Total Monitored Compute Cost	\$86B	\$219B	\$382B
Post-Training/RL	52	104	166
% of Training	79%	79%	79%
Critical-Level Inference	33	114	216
% of Inference	71%	100%	100%
Overhead Compute Cost	\$17B	\$44B	\$76B
% of Monitored Compute	20%	20%	20%
% of Base Compute	15%	18%	18%

Source: Barclays Research, Company Reports

Competitors trailing frontier labs have largely competed on price. Two camps have emerged: #1, the Chinese open-weight providers, which haven't pushed the envelope on monetization, and #2, second-tier western players, largely big-tech companies, willing to subsidize tokens to secure a foothold in the ecosystem. Between the labs and these two cohorts, we see an "efficient frontier" of cost/intelligence forming. At present, less value is captured by these players, but as consumer AI emerges, low-cost providers may capture a greater volume of usage.

FIGURE 25. AI's Emerging Pareto Frontier

Cost Per Task (Log Scale) on the Artificial Analysis Intelligence Index
Source: Barclays Research, Artificial Analysis

China Technology

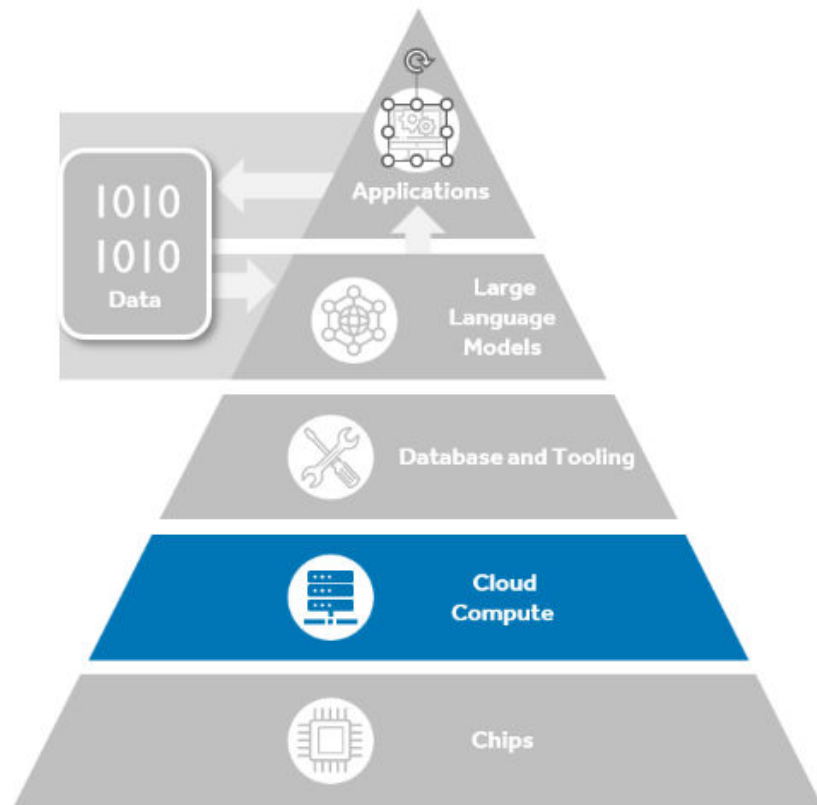
Jiong Shao, CFA BCI, US

Open source is viewed as a national strategy in China, and models can still generate attractive economics despite open-weight and cost-efficient nature. As frontier models scale to hundreds of billions or even trillions of parameters, most enterprises lack the infrastructure required to deploy large MoE models economically. In addition, inference software stacks are increasingly co-optimized with the underlying model architecture, making them difficult for third-party providers to replicate. We believe compute constraints will continue to support model pricing in China by keeping inference capacity structurally tight.

Through our channel checks, MiniMax M3 launched at roughly breakeven gross margin before improving to ~40% within one month through inference optimization alone, with a potential of ~60% margin over the following weeks. DeepSeek reportedly generates 70-80% gross margins on V4, according to The Information, primarily through post training and infrastructure optimization. In the US, Anthropic model is reportedly operating at ~70% gross margin. As such, we believe open-weight models can maintain sustainable inference economics.

AI Cloud Service Providers Delivering the Compute

The cloud service providers, also traditionally known as the hyperscalers (notably Microsoft, AWS, Google and increasingly Oracle), operate large fleets of servers in data centers that they combine with software to deliver multiple cloud services to customers. These vendors fall within the broader category of infrastructure software and benefit from software advancements that have allowed compute to "scale out." Hence, these vendors see strong economies of scale as their hardware and data center investments can be used by thousands of customers who can rent compute and storage from these providers (via OpEx) without spending the significant upfront capital to build/operate a data center and related hardware/networking investments (via CapEx). In recent years, these providers have added purpose-built GPU-powered AI servers to their data centers to address AI training and inference workloads, and have been joined by an emerging cohort of AI focused cloud providers, often referred to as neoclouds (e.g., CoreWeave, Nebius, Crusoe, Lambda and Nscale (Crusoe, Lambda and Nscale all private, not covered)).

FIGURE 26. Cloud Compute Layer in the AI Stack

Source: Barclays Research

Key Risks Driving the Debate

Below, we list some of the key risks underpinning the heightened debate around the emerging AI infrastructure space.

- **Overbuilding concerns and ROI:** The significant capacity agreements (well over \$5tn combined, see [U.S. Semiconductors & Semiconductor Capital Equipment: AI Compute Tracker](#), 9/18/26) amongst major participants could exceed future demand, creating credit risks, and has motivated questions around the return of investment from all the related CapEx needed to service these agreements.
- **Supply chain issues:** There are multiple areas of the AI supply chain that have, and could in the future, see bottlenecks that lead to future issues, mainly around inflation and project timing risks. Given all the demand for AI infrastructure capacity, there is an arms race going on amongst participants that could create supply concerns for critical inputs including power, labor, and DC equipment components, which could then spiral to delays and other issues in more severe cases (cancellations).
- **Questions on useful life of data center equipment:** The AI industry is seeing accelerated innovation cycles that creates questions around the useful life of GPUs used to power AI – as NVIDIA and others product new generations in quicker cycles. Shorter-than-expected useful lives compress the unit economics of players in the AI supply chain (AI compute providers) and could elevate credit risks given usage of debt in the industry. That said, supply shortages and software improvements have helped extend the life of GPUs.
- **"Circular financing" concerns:** There are growing questions around potential circular financing dynamics in the AI ecosystem, where companies supplying critical infrastructure

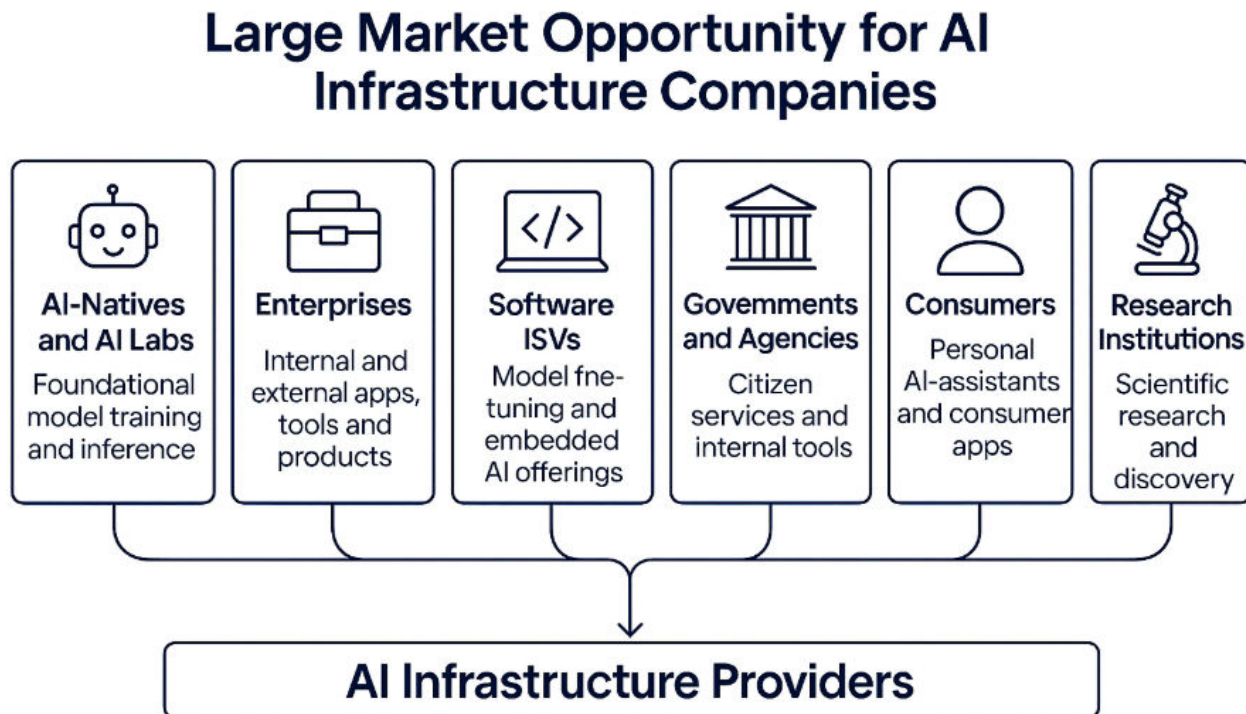
(e.g., GPUs) are investing in AI firms that, in turn, purchase large volumes of those same components. This creates a feedback loop that could artificially inflate demand signals and valuations, as capital flows may be tied to reciprocal business rather than underlying fundamentals. In more extreme cases, this could lead to misallocation of resources and heightened systemic risk if growth expectations fail to materialize.

Aspects Making Us Constructive on the AI Infrastructure Space

In our view, it will be difficult for CY26 to answer all of the debates around the AI industry as there are simply structural aspects, such as lack of visibility into the costs, energy requirements, etc. of future GPU generations, that are impossible to overcome. However, in our view, we are seeing significant progress already on some of these risks (e.g. useful life debate, unit economics amidst supply chain inflation) and the market is overly fixating on single customer risks and not giving appropriate credit to the large, diverse opportunity ahead for AI infrastructure providers. In particular, we see demand as more broad than what recent performance for the space implies, and we see more options around financing and deployment types that could help reduce financing/credit rating downgrade concerns. Put together, we think this creates a compelling long-term opportunity for AI CSPs including the hyperscalers and well positioned neoclouds like DigitalOcean and CoreWeave. We summarize the key areas of the AI infrastructure space that we think are being overlooked.

- AI infrastructure demand is more broad than just model makers and hyperscalers:** We believe we are at the very early innings of enterprise and consumer AI adoption and expect usage to grow exponentially in the coming years. Currently, a significant share of the AI demand is from select AI labs and hyperscalers. However, in the future, this will fall to broader enterprises, organizations, government agencies and consumers. We believe much of the current concerns affecting recent performance stems around the near-term outlook for specific AI labs, with less focus on the larger, long-term enterprise opportunity. We do not find it difficult to underwrite AI usage to see exponential growth in the coming years as AI advancements allow for more machine to machine interactions that drives business processes (e.g., AI agent proliferation, AI frameworks, model routing, discussed previously). Further, we believe the market is g future AI demand from governments and agencies to drive greater efficiency in processes, research better solutions to challenges faced and constituent outcomes.

FIGURE 27. AI Infrastructure Demand is Much More than Just a Few Large Foundational Model Providers



Source: Barclays Research

- AI Unit Economics Described are Compelling:** In our view, the unit economics of the AI infrastructure space are more compelling than what market participants are getting credit for. Remember, not too long ago we did not have a good understanding of the margin profile and unit economics of large AI infrastructure deals. For example, heading into ORCL's October 2025 Analyst Day, conversations discussed AI infrastructure gross margins around ~25% for AI training, with AI inference margins slightly higher, which seemed to be a major overhang on the emerging business model. During ORCL's FAD, the company provided a detailed AI infrastructure contract example with gross margins in the 30-40% range, significantly better than previous expectations. Given these disclosures, below, we detail the unit economics for this example contract, highlighting the compelling return on investment profile of the deal (~16.5% IRR, assuming no debt nor customer prepayment). Assuming debt and customer prepayments to help finance the initial CapEx outlays raises the IRR of the contract. Generally, we note many players in the industry, like Oracle and CoreWeave, only start building AI capacity upon contract signing, which should provide comfort given the positive unit economics built into the original contracts. Hence, as long as the CSP delivers and the customer remains solvent, then these deals should drive healthy economics for the players given the positive unit economics that are embedded in these deals. While future generations of hardware and supply chain inflation is leading to higher upfront CapEx to bring online capacity, the market is seeing higher revenue per MW to offset this, and margin commentary amongst providers have been consistent to better despite this.

FIGURE 28. AI Unit Economics for Example 1GW Contract (6-Year, \$60bn TCV deal)

Example Provided at 2025 Analyst Day (1GW of AI Infra Capacity, 6-Year Contract) Without Leverage								
P&L Summary (\$mn)	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Totals
Revenue		10000	10000	10000	10000	10000	10000	60000
P&L Outlays	-570	-6400	-6400	-6400	-6400	-6400	-6400	-38970
Gross Profit	-570	3600	3600	3600	3600	3600	3600	21030
Breakdown of P&L Costs	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Totals
Land, Data Center and Power	-570	-2240	-2240	-2240	-2240	-2240	-2240	-14010
Compute, Network, Storage		-4160	-4160	-4160	-4160	-4160	-4160	-24960
Breakdown of Cash Costs	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Totals
Land, Data Center and Power	-570	-2240	-2240	-2240	-2240	-2240	-2240	-14010
Compute, Network, Storage	-25000	0	0	0	0	0	0	-25000
Total Cash Costs	-25570	-2240	-2240	-2240	-2240	-2240	-2240	-39010
Project Cash Flow (unlevered)	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Totals
Revenue	0	10000	10000	10000	10000	10000	10000	60000
CapEx + Cash COGS	-25570	-2240	-2240	-2240	-2240	-2240	-2240	-39010
Cash Gross Profit	-25570	7760	7760	7760	7760	7760	7760	20990
OpEx	0	-100	-100	-100	-100	-100	-100	-600
% of revenue	1%	1%	1%	1%	1%	1%	1%	-1%
EBIT	-570	3500	3500	3500	3500	3500	3500	20430
% of revenue		35%	35%	35%	35%	35%	35%	34%
Tax	108	-665	-665	-665	-665	-665	-665	-3882
Effective Tax Rate	19%	19%	19%	19%	19%	19%	19%	19%
Cash Flow (unlevered)	-25462	6995	6995	6995	6995	6995	6995	16508
% of revenue		70%	70%	70%	70%	70%	70%	28%
ROI Calculations (Unlevered)	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Totals
Cumulative Cash Flow	-25462	-18467	-11472	-4477	2518	9513	16508	16508
IRR (unlevered)								16.5%
Payback (Yrs)								3.64
NPV								\$3,883
WACC								10.8%

Based on ORCL's 2025 Financial Analyst Day AI Infrastructure Delivery Example
Source: Company data, Barclays Research

- Cloud Compute Layer Largely AI Model Agnostic:** Throughout this report, we have discussed how the growing AI cost paradox has helped drive a market response to bring down AI costs through AI frameworks and model routing solutions that make it easier to choose the right model for a given task. This dynamic lowers the switching costs for model usage, which creates questions for the compute layer that is powering AI workloads. However, we note that irreverent of the actual AI model being used, AI workloads still require compute, and as outlined earlier, increasingly complex workloads demand greater levels of compute. Hence, we see the compute layer as largely agnostic to the trend of increasing open-source/weight model usage, and see this layer as a key value capture of AI economics as adoption continues.
- AI compute is benefiting from an improved pricing environment:** We note in addition to the healthy unit economics of AI infrastructure, key players in the industry have recently disclosed positive data points around GPU pricing and recontracting rates that suggest an improved pricing backdrop. For example, CoreWeave recently disclosed recently signed short-term contracts with terms of ~3-6 months were signed at an implied ~\$40mn of ARR per

MW, well above the industry's previous rates that were roughly in the low double-digit \$mn ARR per MW range. Further, Oracle during its most recent earnings call noted that GPU utilization remains extremely high (97.9% in fiscal 1Q27) and that capacity that was renewed or resold in the quarter did so at a 20% premium to prior contracts. These trends increase comfort around the inherent financing/leverage risks in the AI infrastructure space.

Well Positioned Vendors

From a higher-level perspective, we see AI cloud service providers as the best positioned subsegment of software to benefit from rising enterprise AI adoption, and note the majority of revenues for this group is directly AI related. The key aspect for investors to consider is that the AI compute layer is largely model agnostic, and hence, this market will see continued growth as AI training and more importantly in the long-run, inference workloads expand as AI adoption continues.

- **Hyperscalers:** Hyperscalers such as AWS, Microsoft Azure, and Google Cloud are well positioned to benefit from AI adoption because they control the foundational layers of the AI stack, including data center infrastructure, networking, storage, and developer services. As enterprise demand for AI continues to outstrip available compute supply, these platforms serve as the primary destination for both AI training and inference workloads. In addition, hyperscalers benefit from a powerful flywheel in which AI consumption drives incremental demand for adjacent cloud services, including databases, data platforms, security, analytics, and application development tools, allowing them to monetize well beyond the compute layer itself.
- **Neoclouds:** Neocloud providers such as CoreWeave are also well positioned because they are purpose-built for AI workloads and can optimize their entire infrastructure stack around GPU-intensive computing. Unlike diversified hyperscalers that balance thousands of services and customer types, neoclouds can focus exclusively on delivering high-performance AI infrastructure with rapid deployment cycles, dense GPU clusters, and specialized networking architectures. As long as AI demand remains well above industry supply and these providers keep offering compelling compute/price performance, these providers can capture workloads from both AI-native companies and large enterprises seeking additional capacity, positioning them as important beneficiaries of the structural expansion in AI compute consumption.

Additional Resources

To learn more about the AI CSP layer within the context of AI adoption, see the below reports.

- [CoreWeave, Inc: New Financing Leans Equity-Based but Pricing Commentary Is the Positive News to Focus on Here](#) (9/17/26)
- [Microsoft Corp.: CQ2 Public Cloud Update; Strong Hyperscaler Results Confirm Healthy AI Demand](#) (7/31/26)
- [DigitalOcean: Healthy GPU Pricing Increases Point to Continued Supply/Demand Imbalance](#) (7/21/26)
- [DigitalOcean: Fundamental Changes Underway: Introducing our New Capacity-Based Model](#) (4/22/26)

China Technology

Jiong Shao, CFA BCI, US

We believe AI cloud providers will remain among the largest beneficiaries of the AI value chain over the next several years, as compute is likely to remain supply constrained, particularly in China. Rising enterprise AI adoption, increasingly agentic workloads, growing consumer usage, and expanding training and inference demand across industries are all driving strong demand for compute infrastructure.

We estimate China's effective frontier training compute is likely less than 5% that of the US after adjusting for chip volume (<20%) and performance. As a result, Chinese cloud providers are aggressively securing compute resources to meet surging AI demand. Capex continues to ramp rapidly across the sector, with Alibaba up 75% yoy to RMB67.7bn and Tencent up 190% yoy to RMB51.8bn in the Jun-Q. Overall, we estimate China big tech capex this year will account for only ~10% of U.S. hyperscaler capex, which Bloomberg consensus forecasts to exceed \$800bn.

Among companies under our China Technology coverage, Alibaba appears particularly well positioned to capture this trend through its full-stack AI cloud platform, spanning proprietary and third-party chips, frontier AI models, MaaS services, and a broad enterprise and consumer ecosystem. Cloud revenue accelerated to 45% yoy in Jun-Q and management guided to ~50% yoy growth in Sep-Q, with further acceleration expected thereafter. It remains confident to achieve \$100bn of external cloud revenue within five years, implying roughly 40% CAGR, while targeting cloud EBITA margins of 20% vs the current low-teens levels. BABA also indicated the current AI infrastructure investments are already generating at least mid-teens ROIC. As utilization rates improve and AI workloads scale, we expect cloud economics to benefit from operating leverage and increasing mix shift toward higher-value AI products and services.

Implications for Powering AI & Digital Infrastructure

Key Risks Driving the Debate

- **The industry faces several converging constraints.** As we've outlined in our Powering AI series, the digital infrastructure buildout faces execution risks across power, capital and skilled labor, alongside increasing local opposition, permitting delays and regulatory fragmentation around data centers. In a compute- and power-constrained world, these bottlenecks increase the value of scarce compute capacity and the electricity required to power it – and are creating urgency for new solutions. Bring-your-own-power (BYOP) can accelerate deployment and reduce cost-shifting concerns, but it is not a panacea (see [Thematic Highlights from Barclays' CEO Energy-Power Conference](#)), as grid expansion remains critical to long-term growth. Constraints are also increasing the focus on compute efficiency, advanced energy technologies and distributed computing architectures, while drawing attention to longer-term possibilities such as space-based data centers.
- **AI may be a national priority, but permission remains local.** Governments increasingly view AI leadership and its supporting infrastructure as strategic national priorities, yet many decisions determining whether and where data centers are built remain state and local. Communities are increasingly focused on electricity costs, water use, land requirements, emissions and other local impacts, creating tension between national AI ambitions and the interests of communities hosting the infrastructure. As we argued in [Powering AI: The Politics of Data Centers](#), community acceptance, permitting and regulatory fragmentation are therefore becoming increasingly important constraints on the pace and location of the buildout.

Aspects of Powering AI & the Digital Infrastructure Buildout

- **AI fundamentally changes the economics of power.** Historically, watts were converted into relatively low-margin physical output ("widgets"); AI increasingly converts watts into high-value, monetizable tokens and intelligence. In our view, this raises the economic value of reliable power and could prove inflationary for power prices, particularly in capacity-constrained markets. A 1 GW AI data-center campus can cost roughly \$50bn, with approximately two-thirds tied to compute infrastructure. Annual compute depreciation alone could exceed \$6bn, versus an annual electricity bill of roughly \$560mn assuming \$0.08/kWh U.S. average industrial rates and 80% utilization. Even dedicated generation at ~\$2-3k/kW would add only ~\$2-3bn to a \$50bn campus. Against that backdrop, power can be relatively inexpensive compared with the value of the compute it enables – and the cost of leaving that compute stranded. We therefore believe hyperscalers are increasingly willing to fund their share of dedicated generation, transmission upgrades and supporting infrastructure to accelerate deployment.
- **AI scaling laws continue to point to rising compute intensity.** Pre-training scaling historically linked performance gains to greater model size, training data and compute. More recently, post-training scaling has increased compute devoted to improving models after initial training, while test-time scaling uses additional inference compute for reasoning and multi-step tasks increasingly important to agentic and physical AI (see [Physical AI: 3 Areas of Humanoid Debate](#)). As AI shifts from single-shot queries toward reasoning, agentic and physical AI workflows, inference demand increasingly reflects more tasks, more tokens per task and, in some cases, more compute per token. These dynamics could more than offset continued model and hardware efficiency gains, supporting higher overall compute and power demand.
- **Open-weight does not mean compute-free.** Routing routine workloads toward smaller or domain-specific models can reduce compute intensity per task, while reserving larger frontier

models for more complex workloads. However, the shift to lower-cost, open-weight models alone does not change the underlying compute or power intensity of a like-for-like workload. Making model weights open can therefore change the economics, ownership and location of compute, but does not by itself reduce the physical requirements for servers, networking, data-center capacity and power.

- **Efficiency could ultimately drive more—not less—compute demand.** Lower token and task costs expand the number of AI use cases capable of generating attractive returns, while broader adoption and increasingly complex workloads increase consumption. This is consistent with the Jevons Paradox, where greater efficiency and lower costs support higher overall consumption. In a compute- and power-constrained world, greater intelligence per GPU or megawatt can therefore increase the productivity of existing infrastructure while expanding the addressable market for AI. The key uncertainty is whether efficiency gains ultimately outpace growth in AI adoption and workload complexity.

Analyst(s) Certification(s):

In relation to our respective sections we, Raimo Lenschow, CFA, Tom O'Malley, William Thompson, Ross Sandler, Saket Kalia, CFA, Hiral Patel, Katherine Ogundiya, Hannah Greenberg, Amy Lian, Laia Marin i Sola, CFA, Jiong Shao, CFA, Dave Dai, CFA and Siddharth Gupta, hereby certify (1) that the views expressed in this research report accurately reflect our personal views about any or all of the subject securities or issuers referred to in this research report and (2) no part of our compensation was, is or will be directly or indirectly related to the specific recommendations or views expressed in this research report.

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Materially Mentioned Stocks (Ticker, Date, Price)

Alibaba Group Holding Ltd. (BABA, 23-Sep-2026, USD 110.80), Overweight/Positive, A/CD/CE/CH/D/E/FA/J/K/L/M/N

Alphabet Inc. (GOOGL, 23-Sep-2026, USD 337.83), Overweight/Positive, A/CD/CE/D/E/J/K/L/M/N

Amazon.com, Inc. (AMZN, 23-Sep-2026, USD 249.27), Overweight/Positive, A/CD/CE/D/E/J/K/L/M/N

Autodesk Inc. (ADSK, 23-Sep-2026, USD 217.15), Overweight/Positive, CD/CE/J

CoreWeave, Inc. (CRWV, 23-Sep-2026, USD 86.90), Equal Weight/Positive, CD/CE/D/FA/J/L

CrowdStrike Holdings, Inc. (CRWD, 23-Sep-2026, USD 262.49), Overweight/Positive, CD/CE/J/K/N

Datadog, Inc. (DDOG, 23-Sep-2026, USD 251.49), Overweight/Positive, CE/J

DigitalOcean (DOCN, 23-Sep-2026, USD 141.21), Overweight/Positive, CE/J

Dynatrace, Inc. (DT, 23-Sep-2026, USD 58.51), Overweight/Positive, CE/J

Elastic N.V. (ESTC, 23-Sep-2026, USD 91.28), Overweight/Positive, CD/CE/J

GitLab Inc. (GTLB, 23-Sep-2026, USD 49.01), Underweight/Positive, FA/J

Microsoft Corp. (MSFT, 23-Sep-2026, USD 500.59), Overweight/Positive, CD/CE/D/FA/J/K/L/M/N

MongoDB, Inc. (MDB, 23-Sep-2026, USD 428.36), Overweight/Positive, CE/J

Oracle Corp. (ORCL, 23-Sep-2026, USD 144.56), Overweight/Positive, CD/CE/J

Palo Alto Networks (PANW, 23-Sep-2026, USD 393.30), Overweight/Positive, CE/D/J/K/L/M/N

Procore Technologies, Inc. (PCOR, 23-Sep-2026, USD 51.06), Overweight/Positive, A/CE/D/E/FA/J/L

Veeva Systems Inc. (VEEV, 23-Sep-2026, USD 270.75), Overweight/Positive, CE/J

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In addition to the stock rating, we provide industry views which rate the outlook for the industry coverage universe as Positive, Neutral or Negative (see definitions below). A rating system using terms such as buy, hold and sell is not the equivalent of our rating system. Investors should carefully read the entire research report including the definitions of all ratings and not infer its contents from ratings alone.

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China Technology

Alibaba Group Holding Ltd. (BABA)	Baidu, Inc. (BIDU)	Bilibili, Inc (BILI)
BYD Company Ltd. (BYDDF)	Contemporary Amperex Technology (CYATY)	Coupang Inc. (CPNG)
Full Truck Alliance Co. Ltd. (YMM)	FUTU Holdings LTD. (FUTU)	Grab Holdings Ltd. (GRAB)
Hon Hai Precision (HNHPF)	JD.com, Inc. (JD)	Kanzhun Ltd. (BZ)
Ke Holdings Inc. (BEKE)	Kuaishou Technology (KUASF)	Li Auto Inc (LI)
Meituan (MPNGF)	NetEase, Inc. (NTES)	Nio Inc (NIO)
PDD Holdings Inc. (PDD)	PONY AI Inc. (PONY)	SEA Ltd. (SE)
Tencent Holdings Ltd. (TCEHY)	Tencent Music Entertainment Group (TME)	Trip.com Group Ltd. (TCOM)
Vipshop Holdings Ltd. (VIPS)	Xiaomi Corporation (XIACY)	XPeng Inc (XPEV)

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Airbnb Inc. (ABNB)	Alphabet Inc. (GOOGL)	Amazon.com, Inc. (AMZN)
Booking Holdings Inc. (BKNG)	Chewy, Inc. (CHWY)	DoorDash, Inc. (DASH)
Duolingo, Inc. (DUOL)	eBay, Inc. (EBAY)	Ethos Technologies Inc. (LIFE)
Etsy Inc (ETSY)	Expedia Inc. (EXPE)	Liftoff Mobile, Inc. (LFTO)
Lyft, Inc. (LYFT)	Maplebear, Inc. (CART)	Match Group, Inc. (MTCH)
Meta Platforms, Inc. (META)	Pinterest, Inc. (PINS)	Roblox Corporation (RBLX)
Shopify (SHOP)	Snap, Inc (SNAP)	Tripadvisor Inc. (TRIP)
Uber Technologies Inc. (UBER)		

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Braze Inc. (BRZE)	CCC Intelligent Solutions (CCC)	Check Point Software Technologies Ltd. (CHKP)
Cloudflare, Inc. (NET)	Commerce (CMRC)	CoreWeave, Inc (CRWV)
CrowdStrike Holdings, Inc (CRWD)	Datadog, Inc. (DDOG)	Descartes Systems Group (DSGX)
DigitalOcean (DOCN)	Dynatrace, Inc. (DT)	Elastic N.V. (ESTC)
EverCommerce Inc. (EVCN)	Five9, Inc. (FIVN)	Fortinet, Inc. (FTNT)
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nCino, Inc. (NCNO)	Okta, Inc. (OKTA)	Omada Health, Inc. (OMDA)
OpenText Corp. (OTEX)	Oracle Corp. (ORCL)	Palo Alto Networks (PANW)
Paycom (PAYC)	Paylocity Holding Corp (PCTY)	Pegasystems, Inc. (PEGA)
Procore Technologies, Inc. (PCOR)	PTC Inc. (PTC)	Rapid7 (RPD)
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